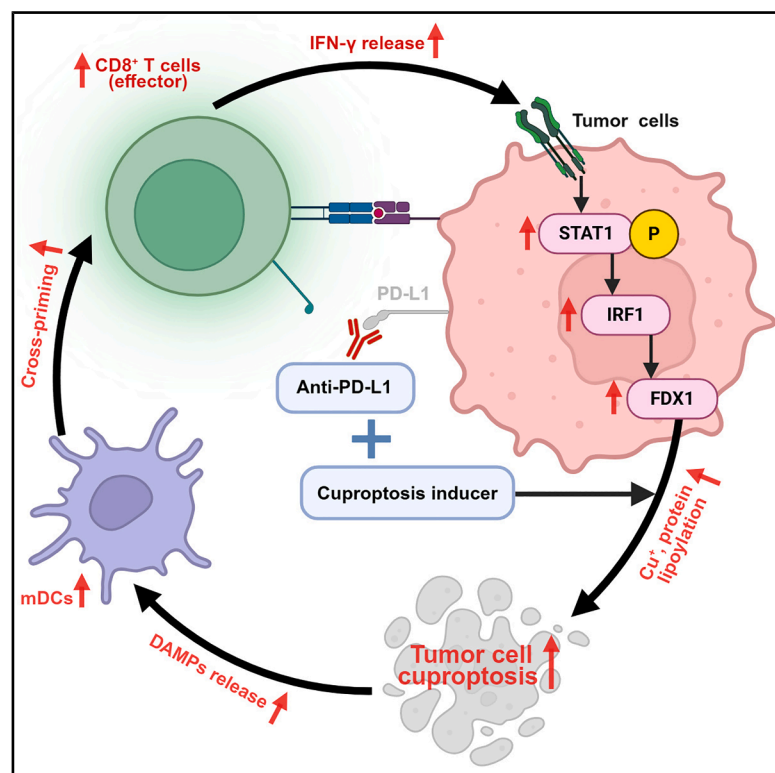


Cuproptosis-immunity crosstalk informs strategy to overcome immunotherapy resistance

Graphical abstract



Authors

Guang Lei, Zhengze Lu, Zhihao Xu, ..., Xingxing Zang, Ziyi Li, Boyi Gan

Correspondence

guanglei_csu@163.com (G.L.),
bgan@mdanderson.org (B.G.)

In brief

A reciprocal circuit links cuproptosis and antitumor immunity, in which CD8⁺ T cell-derived IFN-γ enhances tumor susceptibility to cuproptosis. Engagement of this axis with PD-L1 blockade amplifies tumor cell killing and overcomes immunotherapy resistance.

Highlights

- Cuproptosis acts as immunogenic cell death to activate DCs and cross-prime T cells
- CD8⁺ T cell-derived IFN-γ drives FDX1 transcription to promote tumoral cuproptosis
- Anti-PD-L1 amplifies tumoral cuproptosis and synergizes with cuproptosis inducers
- Cuproptosis induction mitigates intrinsic or acquired resistance to PD-L1 blockade

Article

Cuproptosis-immunity crosstalk informs strategy to overcome immunotherapy resistance

Guang Lei,^{1,12,*} Zhengze Lu,^{1,12} Zhihao Xu,^{1,12} Chen Braun,² David Huo,¹ Jian Gao,³ Lin Tan,⁴ Ting Hong,¹ Shengrong Wu,¹ Mingchuang Sun,¹ Xi Zhao,¹ Qidong Li,¹ Xiong Chen,¹ Yuelong Yan,¹ Hyemin Lee,¹ Chao Mao,¹ Li Zhuang,¹ Li-Ting Ku,⁶ Nahum Puebla,² Hampartsoum Barsoumian,² Jun Yao,³ Lingzhi Hong,⁵ Jianjun Zhang,⁵ Hai Tran,⁵ Jiun-Kae Jack Lee,⁶ Don Gibbons,⁵ Ara Vaporciyan,⁷ John Heymach,⁵ Chunru Lin,³ Eyal Gottlieb,⁸ Mingjian James You,^{9,10} James W. Welsh,² Steven H. Lin,² Xingxing Zang,¹¹ Ziyi Li,⁶ and Boyi Gan^{1,10,13,*}

¹Department of Experimental Radiation Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

²Department of Radiation Oncology, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

³Department of Molecular and Cellular Oncology, Division of Basic Sciences, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

⁴Metabolomics Core Facility, Department of Bioinformatics and Computational Biology, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

⁵Department of Thoracic/Head and Neck Medical Oncology, Division of Cancer Medicine, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

⁶Department of Biostatistics, Division of Discovery Science, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

⁷Department of Thoracic and Cardiovascular Surgery, Division of Surgery, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

⁸Department of Cancer Biology, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

⁹Department of Hematopathology, The University of Texas MD Anderson Cancer Center, Houston, TX 77030, USA

¹⁰The University of Texas MD Anderson Cancer Center UTHealth Graduate School of Biomedical Sciences, Houston, TX 77030, USA

¹¹Department of Microbiology and Immunology, Albert Einstein College of Medicine, Bronx, NY 10461, USA

¹²These authors contributed equally

¹³Lead contact

*Correspondence: guanglei_csu@163.com (G.L.), bgan@mdanderson.org (B.G.)

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SUMMARY

Cuproptosis is a recently identified form of copper-dependent cell death that depends on ferredoxin 1 (FDX1)-mediated protein lipoylation. Here, we reveal that CD8⁺ T cell-mediated antitumor immunity enhances tumor cell susceptibility to cuproptosis, leading to a more potent tumor-suppressive effect of cuproptosis inducers in immunocompetent hosts compared with immunodeficient ones. Mechanistically, cuproptotic tumor cells act as a form of immunogenic cell death, releasing damage-associated molecular patterns that activate dendritic cells and enhance antitumor immunity. Reciprocally, CD8⁺ T cell-derived interferon (IFN)- γ enhances *FDX1* transcription in tumor cells by activating the signal transducer and activator of transcription 1 (STAT1)-IFN regulatory factor-1 (IRF1) signaling axis, resulting in heightened tumor cell sensitivity to cuproptosis. Consequently, combining a cuproptosis inducer with anti-programmed cell death ligand 1 (PD-L1) therapy amplifies tumoral cuproptosis and demonstrates efficacy in overcoming PD-L1 therapy resistance across multiple preclinical models. Our findings unveil a previously unrecognized connection between anti-tumor immunity and cuproptosis and highlight a potential therapeutic approach to counteract tumor immunotherapy resistance by targeting this unique cell death pathway.

INTRODUCTION

Cuproptosis is a recently described form of copper-dependent regulated cell death, triggered by the introduction of copper ions into cells using copper ionophores (such as elesclomol), collectively referred to as cuproptosis inducers.^{1,2} Unlike other forms of regulated cell death such as apoptosis and ferroptosis, the mechanism of cuproptosis primarily involves protein lipoylation—a rare post-translational modifica-

tion occurring in specific mitochondrial metabolic enzymes, such as dihydrolipoamide S-acetyltransferase (DLAT)—mediated by mitochondrial enzymes, including ferredoxin 1 (FDX1).² Excess intracellular copper, as facilitated by copper-loaded ionophores, results in the binding of monovalent copper (Cu⁺) to lipoylated mitochondrial proteins, promoting their oligomerization and depleting iron-sulfur (Fe-S) cluster proteins; these cellular events ultimately lead to cuproptosis (Figure S1A).²

Since its discovery in 2022, cuproptosis has garnered significant interest in cancer research.^{1–4} Although several studies have explored cuproptosis-related nanoparticles in cancer immunotherapy,^{5–7} the role of cuproptosis in shaping antitumor immunity and remodeling the tumor microenvironment (TME) has remained largely unexplored. Antitumor immunity, which is known to regulate various metabolic processes in cancer cells, often promotes metabolic cell death; for example, CD8⁺ T cells, which are key players in antitumor immunity, secrete cytokines such as interferon γ (IFN- γ), which influence tumor cell metabolism and contribute to tumor cell death, including ferroptosis.^{8–10} However, the potential involvement of cuproptosis—a distinct form of metabolic cell death³—in immunity-regulated cell death remains largely unknown. Understanding this interaction could provide insight into mechanisms of CD8⁺ T cell-mediated antitumor immunity and assist in the development of strategies to overcome resistance to immune-checkpoint blockade (ICB).

RESULTS

Cuproptosis inducers exert a stronger tumor-suppressive effect in immunocompetent mice than in immunodeficient mice

To explore the potential roles of the TME and antitumor immunity in cuproptosis-inducing cancer therapy, we implanted multiple mouse cancer cell lines subcutaneously or orthotopically into immunodeficient and syngeneic immunocompetent hosts, followed by treatment with copper-loaded elesclomol or vehicle (Figure S1B). Across all models, copper-loaded elesclomol produced greater tumor suppression in immunocompetent mice than in their immunodeficient counterparts, without affecting body weight (Figures 1A–1D, S1C, and S1D). This effect was abolished by bathocuproinedisulfonic acid (BCS), a Cu⁺ chelator that blocks cuproptosis² (Figures 1E and S1E). Neither elesclomol nor copper alone inhibited tumor growth (Figures S1F and S1G), indicating that their combination is required to induce tumoral cuproptosis.

Cuproptosis depends on protein lipoylation and is marked by Cu⁺-dependent oligomerization and foci formation of lipoylated proteins such as DLAT.² Consistent with the enhanced antitumor effects in immunocompetent mice, baseline tumors in these hosts showed higher levels of lipoylated proteins than did tumors in immunodeficient mice, indicating increased susceptibility to cuproptosis (Figures 1F and 1G). Following treatment, tumors from immunocompetent mice also exhibited higher Cu⁺ signals and DLAT foci (Figures 1H, 1I, and S1H). Another hallmark of cuproptosis onset is the loss of Fe-S cluster proteins, such as succinate dehydrogenase complex iron sulfur subunit B (SDHB).² Consistently, loss of SDHB was more pronounced in tumors from immunocompetent mice (Figures 1J, S1I, and S1J).

Collectively, these results indicate that cuproptosis inducers exert stronger tumor-suppressive effects in immunocompetent mice than in immunodeficient mice, potentially due to the ability of cuproptotic cancer cells to stimulate the immune system and/or the ability of the immune system to further enhance protein lipoylation and cuproptosis susceptibility in tumor cells. We tested these two hypotheses in the following studies.

Cuproptosis as a form of ICD promotes DC maturation and cross-priming

The stronger tumor suppression by cuproptosis inducers in immunocompetent tumors (Figure 1) suggested that tumoral cuproptosis may generate immunostimulatory signals that reshape the TME. Because immunogenic cell death (ICD) engages dendritic cell (DC) cross-presentation and T cell immunity,^{11–14} we tested whether cuproptosis induces canonical ICD features. Cancer cells treated with cuproptosis inducers showed increased levels of damage-associated molecular patterns (DAMPs), including extracellular ATP, HMGB1 release, calreticulin exposure, and IFN- β levels (Figures 2A–2D and S2A–S2D; doxorubicin as a positive control). These increases were also observed in tumors (Figures 2E–2G).

To determine whether cuproptosis qualifies as ICD, we employed a prophylactic vaccination model to test its *in vivo* immunogenicity¹⁵ (Figure 2H). We confirmed that copper-loaded elesclomol-induced death was specifically blocked by BCS but not by other cell death inhibitors (Figure S2E). No tumors developed at the vaccination site, ruling out confounding concomitant immunity driven by residual viable cells. Vaccination with cuproptotic MC38 cells significantly protected against subsequent contralateral tumor challenge and restrained tumor outgrowth, comparable to immunogenic apoptosis and superior to non-immunogenic apoptosis (Figures 2I and 2J), supporting the conclusion that cuproptosis is immunogenic and elicits protective antitumor immunity.

DCs are principal antigen-presenting cells that sense ICD-associated signals and mediate cross-presentation.^{11–14} Cytometry by time-of-flight (CyTOF) analysis in MC38 tumors showed that cuproptosis increased intratumoral major histocompatibility complex class II (MHC II)⁺ DCs and mature DC subsets (Figure 2K). Supernatants from cuproptotic MC38 cells promoted DC maturation marker expression and programmed cell death ligand 1 (PD-L1) upregulation on DCs (Figure S2F) and enhanced DC phagocytosis (Figure S2G), consistent with DAMP-driven DC activation.

DCs in the draining lymph node (dLN) mediate cross-presentation to prime antigen-specific CD8⁺ T cell responses. In dLNs, tumoral cuproptosis increased DC maturation (CD40, CD80, and CCR7) and expanded type 1 conventional DCs (cDC1s), with a concordant rise in PD-L1⁺ DCs (Figure 2L). It also increased DCs presenting H-2K^b-SIINFEKL and expanded tetramer⁺ CD8⁺ T cells (Figures 2M and 2N), supporting enhanced DC-mediated cross-presentation and antigen-specific priming. In line with these antigen-specific responses following cuproptosis inducer treatment, CD8⁺ T cells showed increased activation (CD69⁺), proliferation (Ki67⁺), and cytotoxicity (granzyme B⁺), with enrichment of progenitor exhausted cells and reduction of naive-like cells (Figure 2O). In contrast, non-dLNs showed minimal changes (Figures S2H and S2I).

Finally, DC depletion in CD11c-DTR mice reduced the antitumor effect of cuproptosis induction and abolished expansion of tetramer⁺ CD8⁺ T cells in dLNs (Figures 2P, 2Q, and S2J), demonstrating that DCs are important for cuproptosis-driven T cell responses and tumor control. Collectively, these results suggest that cuproptosis represents a form of ICD that promotes DC maturation, DC-dependent cross-priming, and the expansion of

antigen-specific CD8⁺ T cells and that cuproptosis-mediated tumor suppression depends, at least in part, on DCs.

Tumoral cuproptosis remodels the tumor immune landscape and promotes CD8⁺ T cell effector function

To define how cuproptosis reshapes the tumor immune landscape, we performed CyTOF, flow cytometry, and single-cell RNA sequencing (scRNA-seq; [Figures S2K and S2L](#)) on tumors treated with or without cuproptosis inducers. Consistent with dLN findings ([Figures 2M–2O](#)), intratumoral CD8⁺ T cells showed marker patterns consistent with increased activation, costimulation, and effector-associated differentiation following cuproptosis ([Figures 3A–3D](#), [S2M](#), and [S2N](#)). Tumoral cuproptosis shifted CD8⁺ T cells toward a progenitor- and effector-skewed pattern, expanding progenitor exhausted T cells (T_{PEX}) without increasing terminally exhausted T cells (T_{EX-term}) ([Figure 3E](#)),¹⁶ aligning with increased effector-associated marker expression in CD8⁺ T cells ([Figures 3B–3D](#) and [S2N](#)). Select inhibitory receptor-positive CD8⁺ T cell subsets (e.g., PD-1⁺ or TIM-3⁺) also increased ([Figure 3E](#)), consistent with CD8⁺ T cell activation-associated adaptive inhibitory feedback rather than terminal exhaustion.

Overall, these changes support antigen-specific CD8⁺ T cell activation, in line with enhanced DC-mediated cross-priming in dLNs, while increased inhibitory receptors likely reflect adaptive feedback to strong activation.

Beyond DCs and CD8⁺ T cells, cuproptosis remodeled additional TME components. Activated CD4⁺ T cells increased ([Figures 3A](#) and [3F](#)), whereas natural killer (NK) cells did not increase ([Figure 3A](#)), indicating a T cell-centric response. Tumor-associated macrophages (TAMs) exhibited an increased M1-like/M2-like ratio following cuproptosis induction *in vivo* ([Figures 3A](#) and [3G](#)), with consistent TAM marker changes in scRNA-seq analysis ([Figure S2O](#)), which were recapitulated by conditioned media *in vitro* ([Figure 3H](#)), indicating a shift toward a pro-inflammatory TAM landscape. CSF1R blockade increased the intratumoral M1-like/M2-like TAM ratio¹⁷ ([Figure 3I](#)), enhanced CD8⁺ T cell IFN- γ expression, and improved cuproptosis-mediated tumor control ([Figures 3J–3L](#)), supporting the notion that an M1-like TAM-skewed landscape can promote cuproptosis-mediated tumor suppression.

Collectively, these results suggest that tumoral cuproptosis induces an ICD-like, T cell-centric remodeling of the TME, characterized by DC activation, M1-like TAM polarization, and enhanced T cell effector function.

CD8⁺ T cells and IFN- γ sensitize tumor cells to cuproptosis

Our earlier observations revealed that tumors in immunocompetent mice exhibited enhanced protein lipoylation and cuproptosis susceptibility compared with those in immunodeficient mice ([Figures 1G–1J](#) and [S1J](#)), raising the possibility that anti-tumor immunity might modulate tumor sensitivity to cuproptosis. To test this, we treated tumor-bearing immunocompetent mice with the cuproptosis inducer alongside depleting antibodies targeting specific immune cell populations. Strikingly, CD8⁺ T cell depletion, but not depletion of CD4⁺ T cells or NK cells, significantly attenuated the tumor-suppressive effects of cuproptosis inducer ([Figures 4A](#) and [S3A–S3E](#)).

Upon activation, CD8⁺ T cells secrete several cytokines, including IFN- γ and tumor necrosis factor α (TNF- α), which are critical for their antitumor activity and play key roles in regulating cancer cell death.^{9,10,18} Supernatants from activated human CD8⁺ T cells (treated with anti-CD3/28) enhanced cell death in human cancer cells treated with copper-loaded elesclomol; this enhancement was abolished by the cuproptosis inhibitor BCS, but not by other cell death inhibitors ([Figure 4B](#)), confirming the specific involvement of cuproptosis in this context. Consistently, supernatants from activated CD8⁺ T cells increased Cu⁺ and DLAT foci levels—hallmarks of cuproptosis—under cuproptosis inducer treatment ([Figures 4C–4F](#)).

These effects mediated by CD8⁺ T cells (including increases in Cu⁺ and DLAT foci levels and enhancement of cuproptosis sensitization) were negated by neutralizing IFN- γ but not TNF- α ([Figures 4C–4G](#)), indicating that CD8⁺ T cells promote cancer cell cuproptosis via IFN- γ release. Neutralization of IFN- γ , but not TNF- α , diminished cuproptosis-mediated tumor suppression in immunocompetent mice ([Figures 4H](#) and [S3F–S3H](#)). Depletion of CD8⁺ T cells or neutralization of IFN- γ also reduced lipoylated protein levels in baseline syngeneic tumors ([Figures 4I](#) and [S3I](#)). Similarly, treatment with the cuproptosis inducer increased Cu⁺ and DLAT foci levels while decreasing levels of the Fe-S cluster protein SDHB in syngeneic tumors, whereas depletion of CD8⁺ T cells or neutralization of IFN- γ abolished these effects in syngeneic tumors ([Figures 4J](#), [4K](#), [S3J](#), and [S3K](#)).

Supplementation of IFN- γ *in vitro* sensitized cancer cells to treatment with copper-loaded elesclomol; this sensitization was abolished by the cuproptosis inhibitor BCS, but not by other cell death inhibitors ([Figure 4L](#)), confirming that IFN- γ promotes cuproptosis in cancer cells. The sensitization effect of IFN- γ was more moderate in non-cancer cells (mouse embryonic fibroblasts, MCF10A cells, and primary human CD8⁺ T cells) than in cancer cells ([Figure S3L](#)). Consistently, IFN- γ treatment augmented protein lipoylation and increased elesclomol-copper-induced Cu⁺ signals and DLAT foci formation in cancer cells ([Figures 4M](#) and [S3M–S3P](#)).

In OE33 xenograft tumor models in immunodeficient mice, treatment with copper-loaded elesclomol or IFN- γ alone had marginal or moderate tumor-suppressive effects, while their combination markedly suppressed tumor growth ([Figure 4N](#)). Similar to its effect on protein lipoylation in OE33 cells *in vitro*, IFN- γ treatment also increased protein lipoylation levels in OE33 tumors ([Figure S3Q](#)).

Collectively, these results reveal a significant role of CD8⁺ T cells and IFN- γ in promoting tumoral cuproptosis.

CD8⁺ T cells and IFN- γ promote cancer cell cuproptosis via the upregulation of the STAT1-IRF1-FDX1 axis

We next sought to determine the mechanism by which CD8⁺ T cells and IFN- γ enhance tumor cell susceptibility to cuproptosis. IFN- γ is well known to activate the Janus kinase (JAK)-signal transducer and activator of transcription 1 (STAT1)-IFN regulatory factor-1 (IRF1) signaling pathway, which regulates gene expression and various cellular processes, including cell survival and death in tumor cells.^{8,9,19,20} Tumors from immunocompetent mice exhibited increased levels of STAT1 and IRF1 compared with those from immunodeficient mice; moreover, depletion of

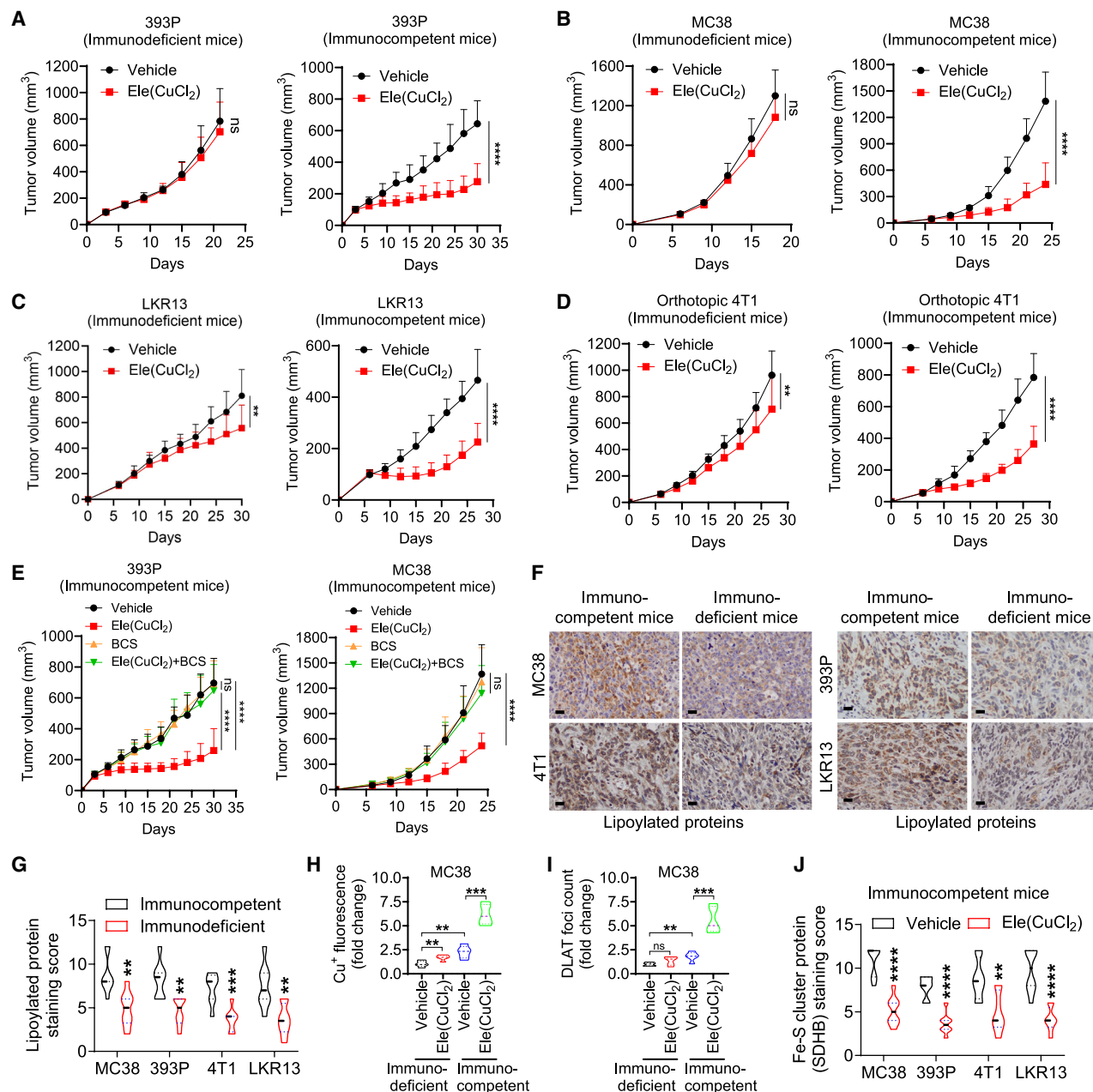


Figure 1. Antitumor immunity promotes tumoral cuproptosis

(A–D) Tumor volumes over time in immunocompetent or immunodeficient mice bearing subcutaneous 393P (A), MC38 (B), or LKR13 (C) tumors, or orthotopic 4T1 tumors (D), treated with elesclomol-copper(II) chloride (CuCl₂) or vehicle.

(E) Tumor volumes over time in immunocompetent mice bearing 393P or MC38 tumors treated with elesclomol-CuCl₂ and/or bathocuproindisulfonic acid (BCS).

(F and G) Representative images (F) or scores (G) of lipoylated protein staining in indicated tumors. Scale bars, 20 μm.

(H and I) Relative Cu⁺ fluorescence intensity (H) or DLAT foci counts (I) in MC38 tumors under indicated treatments.

(J) Fe-S cluster protein SDHB staining scores in indicated tumors from immunocompetent mice treated with elesclomol-CuCl₂ or vehicle.

Data are mean ± SD, n = 6 tumors (A–E). (G–J) Violin plots with median and IQR; n = 5 (H and I) or 8 (G and J) random fields. Unpaired two-tailed t test or two-way ANOVA; *p < 0.05, **p < 0.01, ***p < 0.001, and ****p < 0.0001; ns, not significant.

See also Figure S1.

CD8⁺ T cells or neutralization of IFN-γ significantly reduced STAT1 and IRF1 levels (Figure S4A), confirming that CD8⁺ T cells and IFN-γ activate the STAT1-IRF1 signaling pathway in tumors. RNA-seq

analysis revealed that IFN-γ robustly induced the expression of *STAT1*, *IRF1*, and an array of IFN-stimulated genes (ISGs) under basal and cuproptosis-inducing conditions (Figure S4B; Table S1).

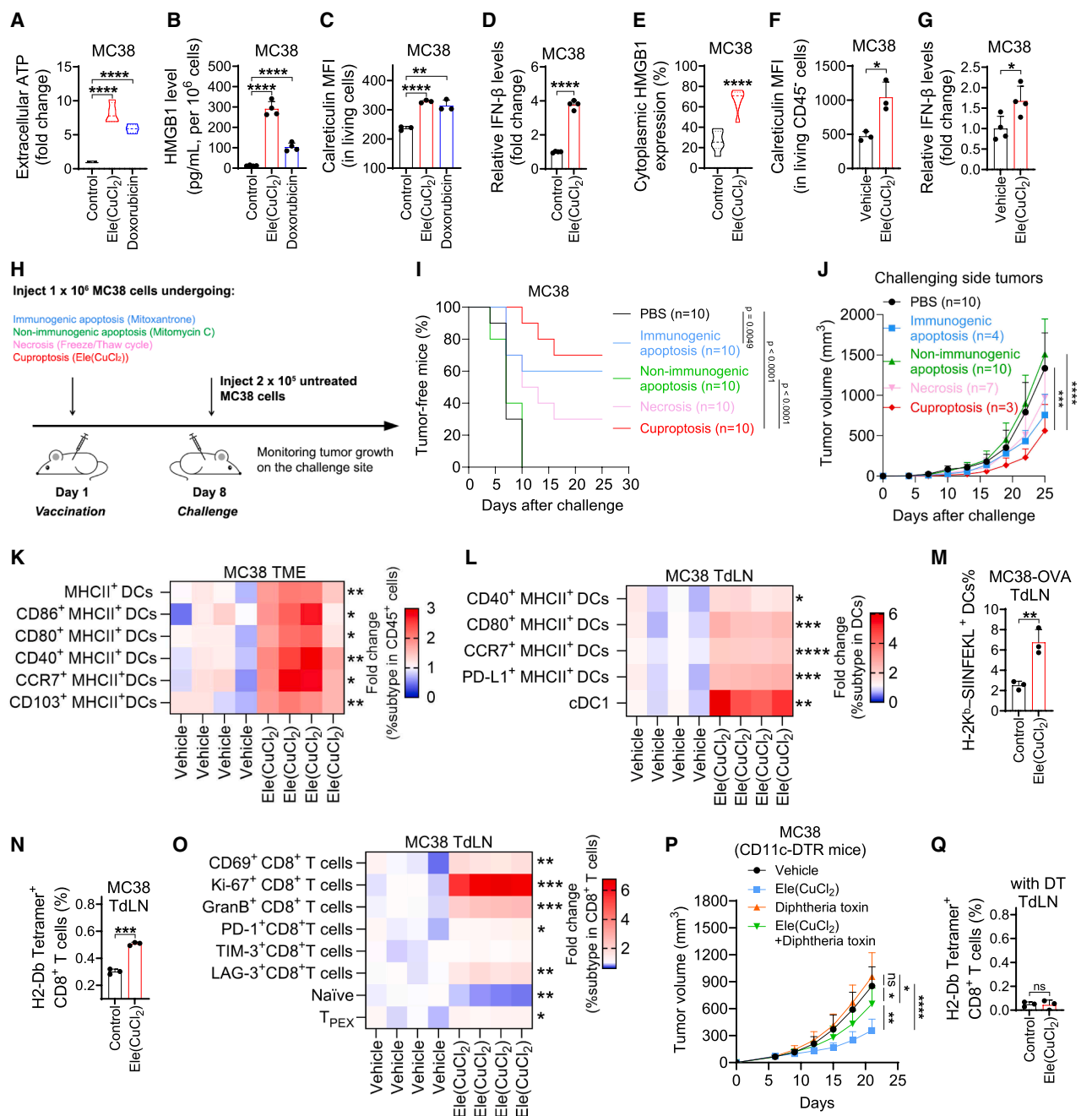


Figure 2. Cuproptosis functions as immunogenic cell death and promotes DC-dependent cross-priming

(A–D) Relative extracellular ATP (A), extracellular HMGB1 (B), ecto-calreticulin mean fluorescence intensity (MFI; C), or relative IFN-β levels (D) in MC38 cells treated with a 2-h pulse of 200 nM elesclomol-2 μM CuCl₂ or 1 μM doxorubicin.

(E–G) Percentage of cytoplasmic HMGB1 expression (E), calreticulin MFI on live CD45⁺ cells (F), or relative IFN-β levels (G) in MC38 tumors treated with elesclomol-CuCl₂ or vehicle.

(H) Prophylactic vaccination scheme using dying MC38 cells induced by immunogenic apoptosis (1 μM mitoxantrone, 24 h), non-immunogenic apoptosis (30 μM mitomycin C, 24 h), necrosis (freeze/thaw, three cycles), or cuproptosis (2-h pulse of 200 nM elesclomol and 2 μM CuCl₂), followed by contralateral challenge with live MC38 cells. PBS served as a no-vaccination control.

(I and J) Tumor-free survival (I) and tumor growth at the challenge site (J) following vaccination. Tumor growth curves in J include only mice that developed tumors after challenge.

(K and L) Fold changes (normalized to vehicle) in frequencies of DC subsets in the MC38 tumor microenvironment (TME; K) or among DCs in tumor-draining lymph nodes (TdLNs; L) under indicated treatments.

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To determine whether enhanced STAT1-IRF1 signaling in syngeneic mouse tumors contributes to CD8⁺ T cell- and IFN- γ -mediated cuproptosis susceptibility, we generated mice bearing control, *STAT1* knockout (KO), and *IRF1* KO syngeneic tumors. *STAT1* or *IRF1* deletion mitigated the tumor growth suppression induced by the treatment with copper-loaded elesclomol (Figures 5A, 5B, and S4C–S4E). These findings underscore the critical role of STAT1-IRF1 signaling activation by CD8⁺ T cell-derived IFN- γ in mediating tumor susceptibility to cuproptosis in immunocompetent mice.

IRF1 is a canonical transcriptional target of STAT1 and acts as a transcription factor to regulate gene expression.^{10,21,22} To explore the mechanism by which IRF1 promotes tumoral cuproptosis in immunocompetent mice, we analyzed the scRNA-seq data from control and *IRF1* KO syngeneic mouse tumors. *IRF1* deletion altered the expression of several genes involved in the cuproptosis pathway in tumor cells, with *FDX1* showing the most pronounced decrease (Figure 5C). Consistently, in OE33 cells, *FDX1* showed the strongest IFN- γ -induced upregulation among known cuproptosis regulators under both basal and cuproptosis-inducing conditions (Figure 5D). Given that *FDX1* is essential for protein lipoylation and cuproptosis,² we have focused on *FDX1* in our subsequent studies.

Tumor cells from immunocompetent mice exhibited higher levels of *FDX1* compared with those from immunodeficient mice, and this increase in *FDX1* expression was abolished by CD8⁺ T cell depletion, IFN- γ neutralization, or the deletion of *STAT1* or *IRF1* (Figures 5E, 5F, and S4F). *FDX1* has a dual role in governing cuproptosis: it converts Cu²⁺ to Cu⁺ and regulates protein lipoylation.^{2,23} Consistently, syngeneic *STAT1* or *IRF1* KO tumors exhibited reduced levels of lipoylated proteins, Cu⁺, and DLAT foci (Figures 5G–5I, S4G, and S4H), mirroring the reduction observed with depletion of CD8⁺ T cells or neutralization of IFN- γ (Figures 4I–4K). Reducing *FDX1* levels in syngeneic tumors abolished cuproptosis-mediated tumor suppression (Figures 5J and S4I). These results suggest an important role of CD8⁺ T cell-derived IFN- γ and the STAT1-IRF1 signaling axis in upregulating *FDX1* expression and protein lipoylation, thereby sensitizing tumor cells to cuproptosis.

We extended our findings to human cancer cells. In OE33 or Hs578T cells, IFN- γ treatment activated the STAT1-IRF1 signaling pathway and increased *FDX1* expression without affecting the levels of other proteins implicated in cuproptosis regulation, such as metal regulatory transcription factor 1 (MTF1), DLAT, and lipoic acid synthetase (LIAS) (Figures 5K, S4J, and S4K). Supernatants from activated human CD8⁺ T cells also upregulated *FDX1* in these cells, and IFN- γ neutralization abrogated this induction (Figures S4L and S4M). Consis-

tent with mild-to-moderate cuproptosis sensitization in non-cancer cells (Figure S3L), IFN- γ only modestly induced *FDX1* expression in these cells (Figures S4N and S4O). Reducing *FDX1* levels attenuated the sensitization of these cells to IFN- γ -mediated cuproptosis, an effect that was reversed by restoration of *FDX1* wild-type but not its enzymatically inactive mutant (Figures 5L and S4P–S4S). Likewise, reducing *FDX1* levels largely abolished the tumor-suppressive effects of combined treatment with the cuproptosis inducer and IFN- γ in OE33 xenograft models (Figures 4N and 5M).

Deletion of *JAK1*, *STAT1*, or *IRF1* significantly diminished IFN- γ -mediated *FDX1* induction and subsequent cuproptosis sensitization (Figures 5N–5Q and S5A–S5F), which was restored by *STAT1* or *IRF1* add-back (Figures S5G–S5J). Conversely, overexpression of a constitutively active *STAT1* (*STAT1-C*)²⁴ or an activity-enhanced *IRF1* mutant (*IRF1-K275, 299R*)²⁵ upregulated *FDX1* and sensitized cells to cuproptosis (Figures S5K–S5P). These findings confirm the essential role of the STAT1-IRF1-*FDX1* axis in mediating IFN- γ -driven sensitization to cuproptosis in human cancer cells.

We next studied the mechanism by which IFN- γ -STAT1-IRF1 signaling regulates *FDX1* expression. Upon IFN- γ -mediated activation of the JAK-STAT1 signaling, IRF1 binds gene promoters, initiating transcription of IFN- γ -responsive genes.^{10,20–22} Analysis of the Encyclopedia of DNA Elements (ENCODE) chromatin immunoprecipitation sequencing (ChIP)-seq database identified binding peaks of IRF1, but not *STAT1*, at an active promoter region within the *FDX1* locus (Figure S5Q). ChIP assays confirmed IRF1 binding to the *FDX1* promoter, with this binding significantly enhanced following IFN- γ treatment (Figures 5R, S5R, and S5S). We cloned the human *FDX1* promoter upstream of a luciferase reporter (*FDX1*^{WT}) and generated a mutant construct (*FDX1*^{mut}) in which the predicted IRF1-binding site was disrupted. Consistently, IRF1 overexpression robustly activated *FDX1*^{WT}, but not *FDX1*^{mut}, promoter luciferase activity (Figures 5S, 5T, and S5T). We further edited the endogenous *FDX1* promoter in near-haploid HAP1 cells and deleted a 46-bp segment encompassing the IRF1 motif (Δ 46-bp cells) (Figure S5U), resulting in reduced IRF1 occupancy on the *FDX1* promoter in Δ 46-bp cells (Figure 5U). We confirmed that deletion of a 46-bp region in the *FDX1* promoter abolished IRF1-induced *FDX1* expression but did not affect *FDX1* expression induced by steroidogenic factor 1 (SF-1)²⁶ (Figures 5V and 5W). Importantly, disruption of the IRF1-binding site on the endogenous *FDX1* promoter did not affect basal *FDX1* levels or elesclomol-copper-induced cuproptosis but specifically abolished IFN- γ -induced *FDX1* expression and cuproptosis sensitization (Figures 5X–5Z). These findings therefore establish *FDX1* as a direct transcriptional target of

(M and N) Frequencies of H-2K^b-SIINFEKL⁺ DCs (M) or H-2D^b tetramer⁺ CD8⁺ T cells (N) in TdLNs of mice bearing MC38-OVA or MC38 tumors under indicated treatments.

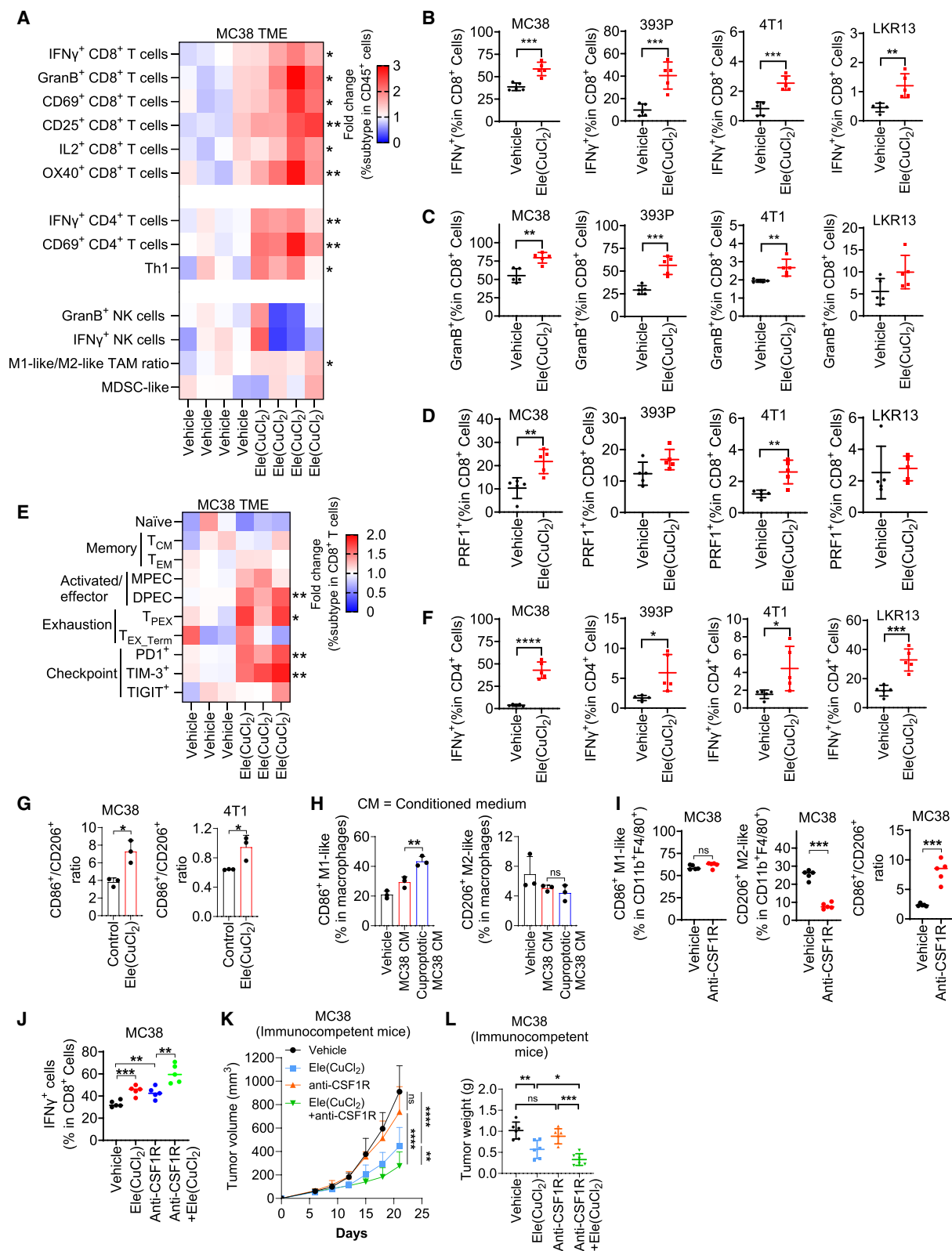
(O) Fold changes in frequencies of CD8⁺ T cell subsets among TdLN CD8⁺ T cells from mice bearing MC38 tumors under indicated treatments.

(P) MC38 tumor growth in CD11c-DTR mice treated with elesclomol-CuCl₂, with or without diphtheria toxin (DT) for DC depletion.

(Q) Frequencies of H-2D^b tetramer⁺ CD8⁺ T cells in TdLNs of DT-treated CD11c-DTR mice bearing MC38 tumors under indicated treatments.

Data are mean $\pm$ SD, $n = 3$ –4 independent repeats (B–D), 3–6 tumor or TdLN samples (F, G, M, N, P, Q), and 3–10 tumors (J). (I) Kaplan-Meier survival curve ($n = 10$ mice). (A and E) Violin plots with median and IQR; $n = 4$ independent repeats (A) or 6 random fields (E). (K, L, and O) Heatmaps; $n = 4$ tumor (K) or TdLN (L and O) samples. Unpaired two-tailed t test, two-way ANOVA, or log-rank test; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, and **** $p < 0.0001$; ns, not significant.

See also Figure S2.



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IRF1 and support a critical role of IRF1-mediated *FDX1* transcription in IFN- γ -driven *FDX1* upregulation and downstream cuproptosis.

FDX1 is known to support mitochondrial respiration and tricarboxylic acid (TCA) cycle activity, likely because it is required for the protein lipoylation of key TCA cycle enzymes.² In line with *FDX1* upregulation, IFN- γ treatment increased oxygen consumption rate, elevated TCA cycle-associated metabolites, and decreased glycolysis-associated metabolites (Figures S6A–S6D), resulting in a metabolic profile that favors cuproptosis induction. Consistently, inhibition of electron transport chain (ETC) complex I or complex III partially attenuated IFN- γ -induced sensitization to cuproptosis (Figure S6E). These results support a model in which IFN- γ increases cuproptosis susceptibility, at least in part, through *FDX1*-supported mitochondrial oxidative metabolism. We acknowledge that *FDX1*-independent mechanisms might also contribute to enhanced mitochondrial respiration, because IFN- γ also increased ETC gene expression (Figure S6F).

Finally, we assessed the clinical relevance of the STAT1-IRF1-*FDX1* axis. Across many cancer types in The Cancer Genome Atlas (TCGA), *FDX1* expression positively correlated with the levels of STAT1, IRF1, or multiple ISGs (e.g., CXCL9/10/11), as well as the CD8⁺ T cell activation signature (Figure S6G). Similar patterns were observed in GEO datasets for lung, colon, esophageal, and breast cancers (Figure S6H). Additionally, in a tissue-microarray cohort of lung cancer patients, we observed positive correlations between *FDX1* and IRF1/STAT1 (Figures S6I and S6J). These human data align with our mechanistic model.

Together, our data suggest that CD8⁺ T cell-secreted IFN- γ activates the STAT1-IRF1 signaling pathway, driving *FDX1* transcription and thus sensitizing tumor cells to cuproptosis.

Anti-PD-L1 therapy boosts tumor susceptibility to cuproptosis induction

Our aforementioned data suggest a reciprocal regulation between tumoral cuproptosis and CD8⁺ T cell-mediated antitumor immunity: on one hand, tumoral cuproptosis, as a form of ICD, enhances CD8⁺ T cell function. On the other hand, CD8⁺ T cell-secreted IFN- γ in turn promotes tumoral cuproptosis. These observations prompted us to further investigate whether boosting antitumor immunity with ICB could further amplify tumoral cuproptosis. ICB, such as anti-PD-1/PD-L1 and anti-cytotoxic T-lymphocyte-associated protein 4 (CTLA-4), are well established, standard-of-care immunotherapies in clinical set-

tings.²⁷ We noted that *in vivo* treatment with copper-loaded elesclomol significantly increased PD-L1 levels in tumors (Figure 6A), consistent with another study revealing that copper supplementation elevated PD-L1 expression.²⁸ We hypothesized that this upregulation of PD-L1 might represent an adaptive response of tumor cells to counteract the antitumor effects of cuproptosis inducer treatment; consequently, combining anti-PD-L1 therapy with cuproptosis inducers could further enhance their tumor-suppressive effects.

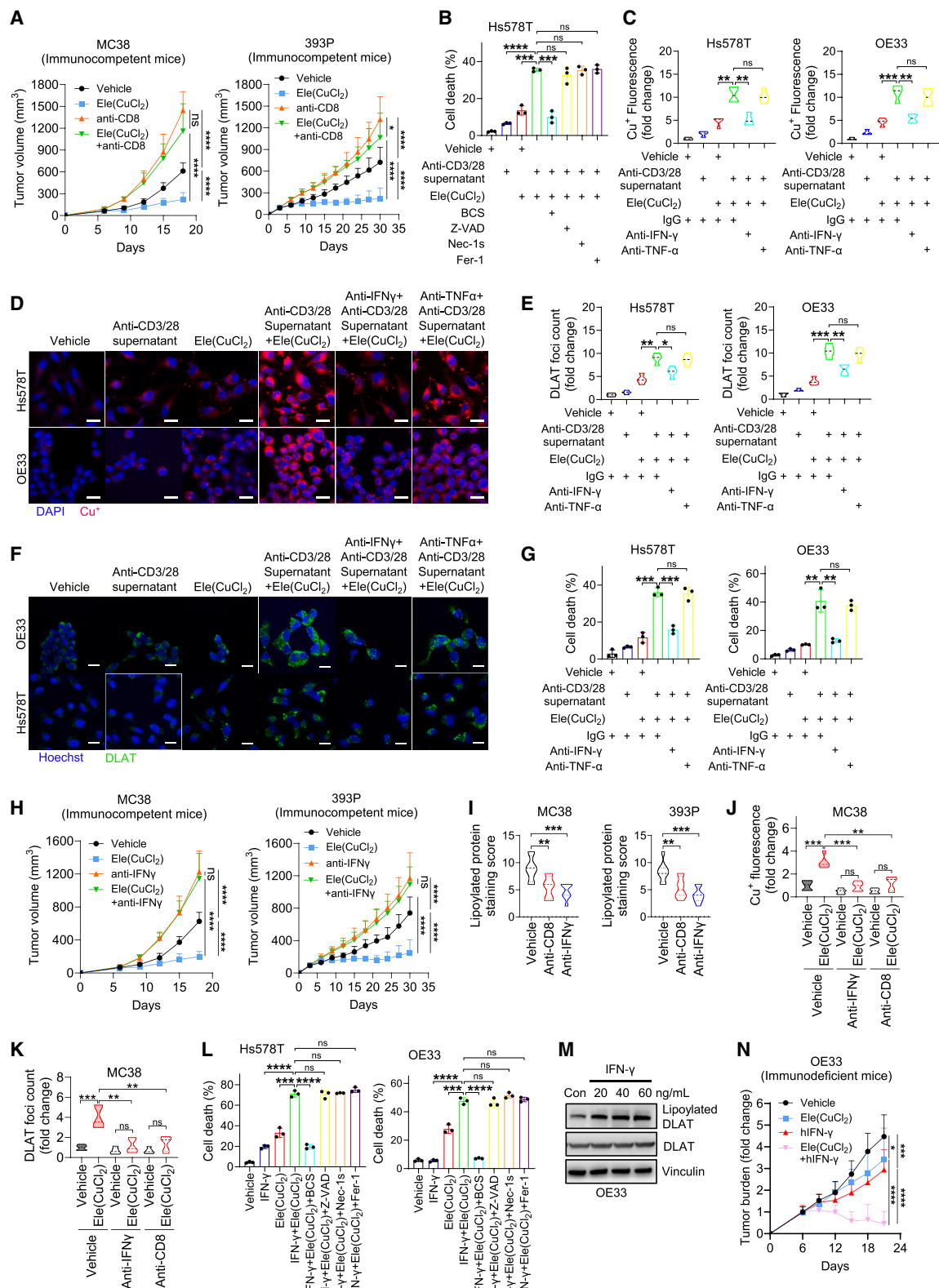
Supporting this hypothesis, the combination of a PD-L1 antibody with copper-loaded elesclomol nearly halted MC38 tumor growth in syngeneic models (Figure 6B). In our animal studies thus far, we have used copper chloride-loaded elesclomol as the cuproptosis inducer for *in vivo* treatment. To facilitate the clinical translation of our therapeutic strategy, we further tested the combination of anti-PD-L1 with elesclomol + copper gluconate, a clinically available copper formulation used as a nutrient supplement.²⁹ Remarkably, this combination therapy nearly eradicated the growth of subcutaneous MC38 and orthotopic EMT6 tumors in syngeneic models (Figures 6C and 6D). Bliss independence model analysis suggested that the combination effects are synergistic (Figure 6E).

Consistently, copper gluconate + elesclomol increased PD-L1 levels and increased the proportion of IFN- γ ⁺ CD8⁺ or CD4⁺ T cells (Figures 6F–6H). Accordingly, combining anti-PD-L1 with copper gluconate + elesclomol further enhanced intratumoral IFN- γ ⁺ T cell responses (Figures 6G and 6H). These findings suggest that, in the context of cuproptosis inducer treatment, blocking cuproptosis-induced PD-L1 further increases IFN- γ ⁺ T cell frequencies within tumors, thereby promoting tumoral cuproptosis.

In line with the above observations, this combination therapy significantly increased IRF1 levels and *FDX1* mRNA expression in tumors (Figures 6I and 6J). Although *FDX1*-mediated protein lipoylation is required for cuproptosis, both *FDX1* and protein lipoylation have also been shown to be depleted during this cell death, likely because *FDX1* itself is a Fe-S cluster protein.² (This depletion represents a distinctive yet perplexing feature of cuproptosis, as *FDX1* depletion is expected to confer resistance to cuproptosis. However, it is important to note that *FDX1* depletion is a *consequence*, not a *cause*, of cuproptosis. By the time *FDX1* is depleted due to Fe-S cluster protein loss, the downstream program triggering cuproptosis has likely already been initiated, rendering the process irreversible.) Consistently, we observed decreased levels of *FDX1* and protein

Figure 3. Tumoral cuproptosis reshapes the tumor microenvironment and enhances CD8⁺ T cell effector function

(A) CyTOF analysis showing fold changes in frequencies of indicated immune cell subsets in the MC38 TME following elesclomol-CuCl₂ treatment. (B–D) Proportions of IFN- γ ⁺ (B), granzyme B⁺ (C), or perforin⁺ (D) cells among CD8⁺ T cells in MC38, 393P, 4T1, or LKR13 tumors under indicated treatments. (E) Fold changes in frequencies of indicated subsets among CD8⁺ T cells in the MC38 TME following elesclomol-CuCl₂ treatment. (F) Proportion of IFN- γ ⁺ cells among CD4⁺ T cells in MC38, 393P, 4T1, or LKR13 tumors under indicated treatments. (G) CD86⁺/CD206⁺ TAM ratio in MC38 or 4T1 tumors under indicated treatments. (H) Frequencies of CD86⁺ or CD206⁺ cells among M0 macrophages treated with vehicle control or conditioned medium (CM) from live or cuproptotic MC38 cells. (I and J) Frequencies of CD86⁺ or CD206⁺ TAMs (I), the CD86⁺/CD206⁺ TAM ratio (I), or the proportion of IFN- γ ⁺ CD8⁺ T cells (J) in MC38 tumors under indicated treatments. (K and L) MC38 tumor volume (K) and weight (L) under indicated treatments. Data are mean $\pm$ SD, $n = 3$ –6 tumors or tumor samples (B–D, F, G, and I–L) or 3 independent repeats (H). (A and E) Heatmaps ($n = 3$ –4 tumor samples). Unpaired two-tailed *t* test or two-way ANOVA; **p* < 0.05, ***p* < 0.01, ****p* < 0.001, and *****p* < 0.0001; ns, not significant. See also Figure S2.



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lipoylation in tumors treated with copper gluconate + elesclomol (Figure 6I); however, treatment with anti-PD-L1 alone, or in combination with copper gluconate + elesclomol, significantly increased FDX1 and protein lipoylation levels (Figure 6I), correlating with the upregulated *FDX1* mRNA expression (Figure 6J) and enhanced tumor-suppressive effect of the combination therapy. In line with these changes, the combination therapy enhanced Cu⁺ signals and DLAT foci formation—markers of cuproptosis—in tumors (Figures 6K and 6L).

Collectively, our findings suggest a bidirectional crosstalk between antitumor immunity and cuproptosis. In this circuit, CD8⁺ T cell-derived IFN- γ upregulates FDX1 transcription via the STAT1-IRF1 axis, thereby sensitizing tumor cells to cuproptosis; in turn, tumoral cuproptosis activates DCs and enhances CD8⁺ T cell-dependent antitumor immunity, while increasing tumoral PD-L1, potentially curbing its antitumor effects (Figure 6M). Accordingly, combining anti-PD-L1 with a cuproptosis inducer synergistically increases intratumoral T cell activity, elevates FDX1 expression in tumor cells, and amplifies tumor cuproptosis, thereby augmenting this reciprocal interaction (Figure 6N).

Inducing tumoral cuproptosis mitigates immunotherapy resistance

Despite the significant success of ICB in cancer therapy, many patients experience intrinsic or acquired resistance, limiting the overall efficacy of these treatments.^{30–32} A major contributor to ICB resistance is a “cold” TME, with poor tumor immunogenicity, downregulation of antigen presentation, limited baseline T cell infiltration, and myeloid-dominated suppression.^{33–39} Our model presented above (Figures 6M and 6N) indicates that tumoral cuproptosis could remodel the TME toward a more immune-inflamed (“hotter”) state, providing a mechanistic rationale for its potential synergy with anti-PD-L1 therapy in overcoming ICB resistance. We therefore examined whether inducing tumoral cuproptosis could help reverse this resistance.

To evaluate whether inducing cuproptosis can overcome intrinsic resistance to immunotherapy, we used the spontaneous MMTV-PyMT mammary tumor model and established subcutaneous or orthotopic syngeneic tumor models from several tumor cell lines known for their ICB resistance, including lung can-

cer (393P and LLC), breast cancer (4T1), and pancreatic cancer (PAN02). As expected, those tumors exhibited intrinsic resistance to anti-PD-L1 treatment; importantly, the combination of anti-PD-L1 therapy with copper gluconate + elesclomol significantly suppressed the growth of both spontaneous MMTV-PyMT tumors and syngeneic tumors, and markedly reduced lung metastases in MMTV-PyMT mice (Figures 7A–7F and S7A). Bliss independence model analysis suggested that the combination effects were synergistic across all tested ICB-resistant models (Figure S7B). In addition, hypoxia/hypoxia-inducible factor 1 alpha (HIF-1 α) has been reported to modulate tumoral cuproptosis and responses to ICB.^{40,41} Although neither CD8⁺ T cells nor IFN- γ altered HIF-1 α levels (Figures S7C and S7D), HIF-1 α inhibitor PX-478 further enhanced the antitumor efficacy of cuproptosis inducer + anti-PD-L1 combination in pancreatic tumors characterized by pronounced hypoxia⁴² (Figure S7E).

In our previous study,³³ we generated syngeneic tumors from 344SQ parental cells and then subjected them to continuous anti-PD-1 treatment; through multiple rounds of selection, we established 344SQ-resistant (344SQ_R) cells, which had acquired resistance to both anti-PD-1 and anti-PD-L1 therapies. We next investigated whether inducing cuproptosis could overcome acquired resistance to ICB using the 344SQ_R syngeneic model. As expected, 344SQ_R tumors exhibited complete resistance to anti-PD-L1 treatment; treatment with copper gluconate + elesclomol alone moderately suppressed tumor growth, while its combination with anti-PD-L1 therapy substantially inhibited the growth of 344SQ_R tumors (Figures 7G and S7A). These results suggest that cuproptosis induction can mitigate intrinsic or acquired resistance to immunotherapy in multiple tumor models.

Consistent with our earlier observations in other models (Figures 2A–2G and S2A–S2D), 344SQ_R and 4T1 cells or tumors treated with cuproptosis inducers exhibited ICD hallmarks, including increased calreticulin exposure, HMGB1 and ATP release, and IFN- β induction (Figures 7H and S7F–S7I). Further analyses revealed that cuproptosis induction in these resistant tumors increased frequencies of MHC II⁺ DCs and their maturation marker-positive subsets, raised the M1-like/M2-like TAM ratio, reduced myeloid-derived suppressor cell (MDSC)-like proportions, and expanded activated CD4⁺ and CD8⁺ T cell subsets

Figure 4. CD8⁺ T cells and IFN- γ sensitize tumor cells to cuproptosis

(A) Tumor volumes over time in immunocompetent mice bearing MC38 or 393P tumors treated with elesclomol-CuCl₂ and/or anti-CD8 depleting antibodies. (B) Cell death in Hs578T cells treated with supernatant from anti-CD3/CD28-stimulated CD8⁺ T cells (48 h) and/or 2-h pulse of elesclomol-CuCl₂, with or without 40 μ M BCS, 5 μ M Z-VAD, 2 μ M Nec-1s, or 5 μ M ferrostatin-1. (C–G) Relative Cu⁺-probe fluorescence intensity (C), representative Cu⁺-probe staining images (D), relative DLAT foci quantification (E), representative DLAT staining images (F), or cell death (G) in Hs578T or OE33 cells treated with supernatant from anti-CD3/CD28-stimulated CD8⁺ T cells (48 h) and/or 2-h pulse of elesclomol-CuCl₂, with or without 10 μ g/mL anti-IFN- γ or anti-TNF- α blocking antibody. Scale bars, 20 μ m. (H) Tumor volumes over time in immunocompetent mice bearing MC38 or 393P tumors treated with elesclomol-CuCl₂ and/or anti-IFN- γ blocking antibodies. (I–K) Lipoylated protein staining scores (I), relative Cu⁺ fluorescence intensity (J), or relative DLAT foci counts (K) in MC38 (I–K) or 393P (I) tumors from immunocompetent mice under indicated treatments. (L) Cell death in Hs578T or OE33 cells treated with IFN- γ (48 h) and/or 2-h pulse of elesclomol-CuCl₂, with or without 40 μ M BCS, 5 μ M Z-VAD, 2 μ M Nec-1s, or 5 μ M ferrostatin-1. (M) Lipoylated and total DLAT levels in OE33 cells treated with IFN- γ at indicated concentrations for 24 h. (N) Relative tumor burden over time in OE33 xenografts in immunodeficient mice treated with IFN- γ and/or elesclomol-CuCl₂. Data are mean $\pm$ SD, n = 6–8 tumors (A, H, and N) or 3 independent repeats (B, G, and L). (C, E, and I–K) Violin plots with median and IQR; n = 4–8 random fields. Unpaired two-tailed t test or two-way ANOVA; * p < 0.05, ** p < 0.01, *** p < 0.001, and **** p < 0.0001; ns, not significant. See also Figure S3.

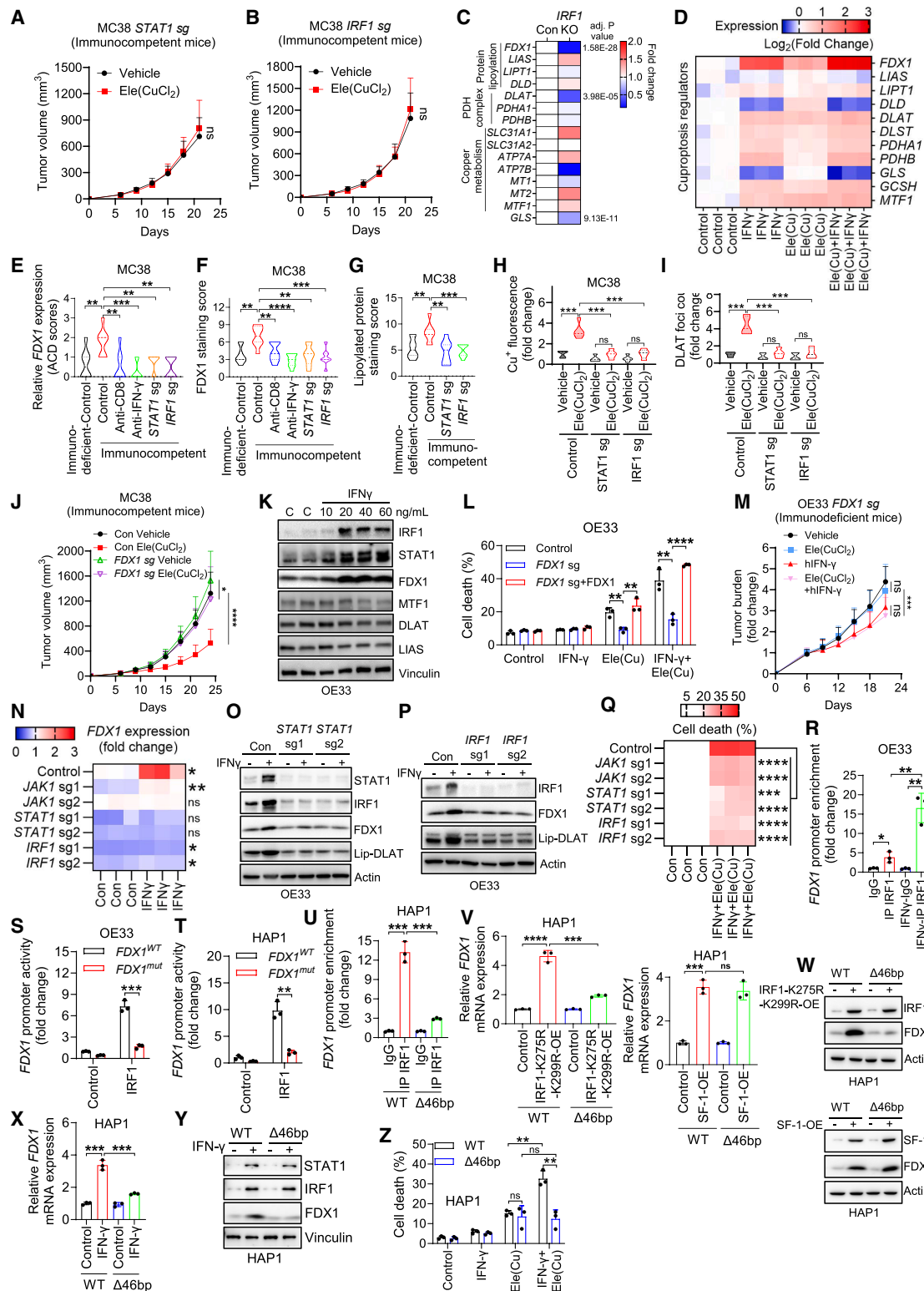


Figure 5. CD8⁺ T cell-IFN-γ promotes cuproptosis via upregulating the IRF1-FDX1 axis

(A and B) Tumor volumes over time in immunocompetent mice bearing *STAT1*- or *IRF1*-sgRNA-infected MC38 tumors treated with elesclamol-CuCl₂.

(C and D) Fold changes in the expression of cuproptosis-regulatory genes in control versus *IRF1*-deficient tumor cells from MC38 tumors in immunocompetent mice (C) or in OE33 cells under the indicated treatments (D).

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(Figures 7I and S7J). Multiplex cytokine profiling aligned with these immune shifts, showing upregulation of DC/Th1/CD8⁺ T cell-associated cytokines and a higher IFN- γ /interleukin 10 (IL-10) ratio, along with reduced immunosuppressive mediators (Figure 7J).

We observed that treatment with copper gluconate + elesclomol increased tumoral PD-L1 levels (Figure S7K); correspondingly, the combination therapy produced a more pronounced increase in the proportions of activated/effector-marker-positive T cells than either monotherapy in these ICB-resistant tumor models (Figures 7K, 7L, and S7L). In these tumors, compared with the cuproptosis inducer alone, the combination therapy further increased the levels of IRF1, FDX1, protein lipoylation, Cu⁺ signals, and DLAT foci (Figures 7M–7Q and S7M–S7P), consistent with our previous observations (Figures 6I–6L). Importantly, these treatments did not cause significant toxicity, including hematological, hepatic, cardiac, splenic, or renal toxicity, nor did they induce weight loss in animal studies (Figures S8A–S8C). These findings highlight the favorable tolerability of this combination therapy *in vivo*.

Finally, analyses of samples or sequencing data from patients revealed that, among ICB-treated patients, tumors from responders exhibited higher FDX1 expression than those from non-responders, and FDX1-high status was associated with longer survival, implicating FDX1 as a candidate biomarker associated with ICB response (Figures S8D–S8G).

Collectively, these findings from diverse preclinical models and clinical samples underscore the potential of inducing cuproptosis as a therapeutic strategy to counteract resistance to immunotherapy.

DISCUSSION

The intersection of cell death pathways with antitumor immunity has repeatedly reshaped our understanding of cancer biology, exemplified by ferroptosis research in recent years. Initially defined as a tumor-intrinsic metabolic vulnerability,⁴³ ferroptosis was later shown to be promoted by CD8⁺ T cells and to synergize with ICB,⁹ repositioning it within the broader immune-tumor

ecosystem and substantially expanding the field.⁴⁴ Analogously, our study establishes cuproptosis as an immunologically integrated cell death pathway. We show that cuproptotic tumor cells release DAMPs that activate DCs and enhance antitumor immunity, while T cell-derived IFN- γ sensitizes tumor cells to cuproptosis, forming a reciprocal immunometabolic feedback loop (Figures 6M and 6N). Prior studies demonstrate that cuproptosis-associated mitochondrial injury can trigger mtDNA release and activate cyclic GMP–AMP synthase (cGAS)–stimulator of interferon genes (STING) signaling,^{45,46} consistent with our observation of type I IFN induction. Together, these findings position cuproptosis as a key interface linking tumor metabolism, immune activation, and therapeutic response.

More broadly, emerging mitochondria-centered cell death pathways, such as mitoxiperiosis,⁴⁷ further highlight how mitochondrial metabolic state shapes immune signaling and cell fate. Although mechanistically distinct, these pathways converge on a common principle: mitochondria act not only as metabolic hubs but also as integrators of immune and stress signals that govern tumor immunogenicity and therapy response.

Our study also has broad implications for overcoming ICB resistance. Building on this reciprocal circuit, we propose that inducing tumoral cuproptosis can shift immunologically “cold” or immune-excluded tumors toward a more inflamed state by restoring DC-dependent cross-priming and rebalancing the myeloid compartment toward a higher M1-like/M2-like TAM ratio. This remodeling expands activated tumor-reactive CD8⁺ T cells and is associated with stronger effector-associated cytokine and cytotoxic marker programs, potentially re-establishing PD-1/PD-L1 as an important adaptive checkpoint and thereby restoring sensitivity to PD-1/PD-L1 blockade.

When intracellular copper levels become excessive, Cu⁺ binds to lipoylated proteins, triggering their oligomerization and the depletion of Fe-S cluster proteins; this cascade induces acute proteotoxic stress, culminating in the onset of cuproptosis.² In *FDX1*-deficient cells, impaired protein lipoylation effectively suppresses cuproptosis.² Notably, during cuproptosis, the depletion of Fe-S cluster proteins includes the loss of FDX1 itself,

(E–G) Relative *FDX1* mRNA levels by advanced cell diagnostics (ACD) score (E), or immunohistochemical staining scores for FDX1 (F) or lipoylated proteins (G) in control, *STAT1*[−], or *IRF1*-sgRNA-infected MC38 tumors from immunodeficient or immunocompetent mice under indicated treatments.

(H and I) Relative Cu⁺ fluorescence intensity (H) or DLAT foci levels (I) in the control, *STAT1*[−], or *IRF1*-sgRNA-infected MC38 tumors from immunocompetent mice under indicated treatments.

(J) Tumor volumes over time in immunocompetent mice bearing Cas9 control or *FDX1*-sgRNA-infected MC38 tumors treated with elesclomol-CuCl₂ or vehicle.

(K) Levels of indicated proteins in OE33 cells treated with IFN- γ at indicated concentrations for 24 h.

(L) Cell death in OE33 cells with Cas9 control, *FDX1*-sgRNA infection, or *FDX1*-sgRNA infection restored with FDX1 after IFN- γ treatment (48 h) and/or a 2-h pulse of elesclomol-CuCl₂.

(M) Relative tumor burden over time in *FDX1*-sgRNA-infected OE33 xenografts in immunodeficient mice treated with IFN- γ and/or elesclomol-CuCl₂.

(N–Q) Relative *FDX1* mRNA levels (N), indicated protein levels (O and P), or cell death (Q) in Cas9 control, *JAK1*[−], *STAT1*[−], or *IRF1*-sgRNA-infected OE33 cells treated with IFN- γ , with or without a 2-h pulse of elesclomol-CuCl₂.

(R) ChIP-qPCR showing the binding of IRF1 to the *FDX1* promoter region in OE33 cells treated with or without IFN- γ .

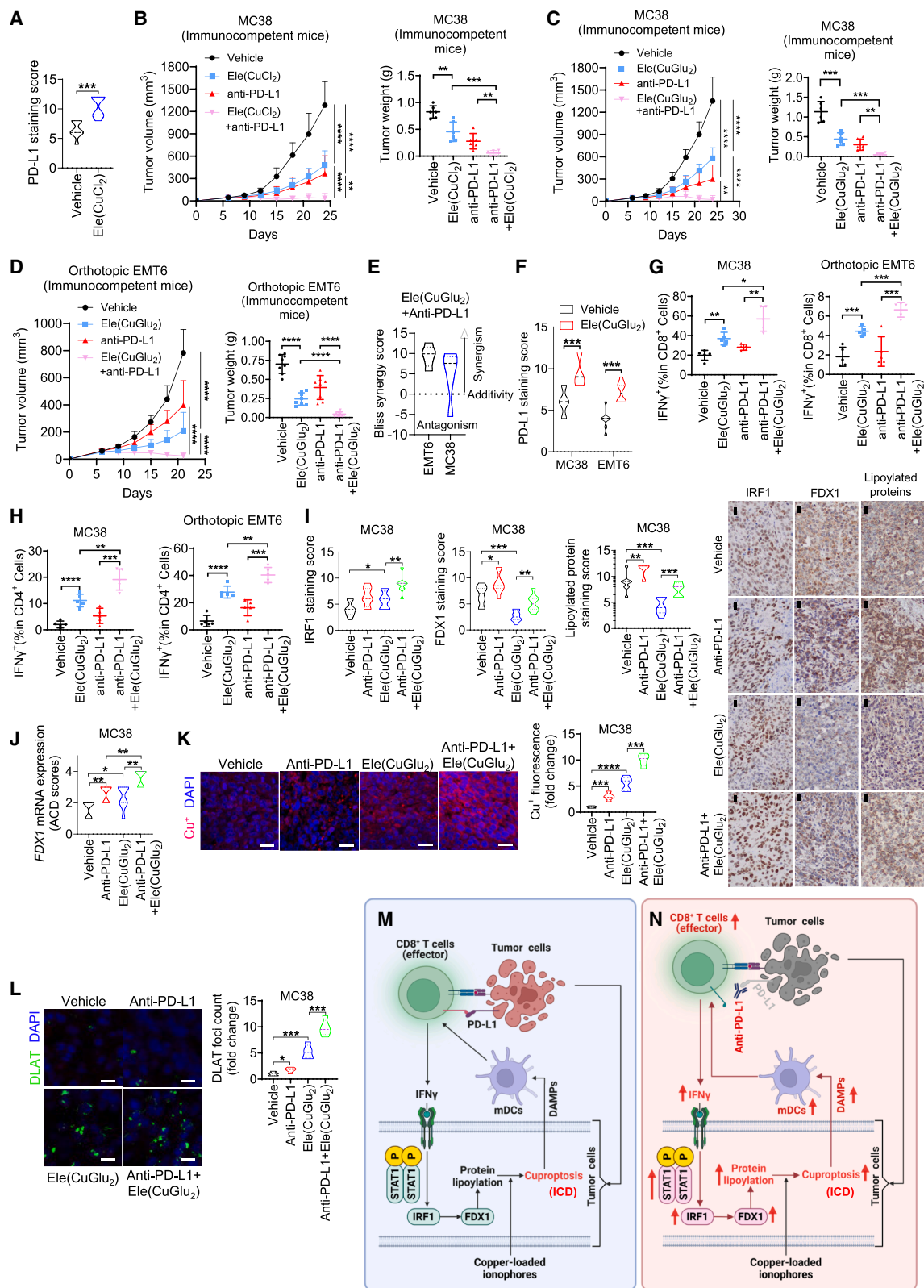
(S and T) WT or IRF1-motif-deleted *FDX1* promoter reporter activity in OE33 (S) or HAP1 (T) cells with IRF1 overexpression.

(U–W) IRF1 ChIP at the *FDX1* promoter (U), *FDX1* mRNA levels (V), or indicated protein levels (W) in WT or Δ 46-bp HAP1 cells of indicated genotypes.

(X–Z) *FDX1* mRNA levels (X), indicated protein levels (Y), or cell death (Z) in WT or Δ 46-bp HAP1 cells under indicated treatments.

Data are mean $\pm$ SD, $n = 8$ tumors (A, B, J, and M) or 3 independent repeats (L, R–V, X, and Z). (E–I) Violin plots with median and IQR; $n = 5$ (H and I) or 8 (E–G) random fields. (D, N, and Q) Heatmaps ($n = 3$ independent repeats). Unpaired two-tailed *t* test, two-way ANOVA, or Wilcoxon rank-sum test; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; ns, not significant.

See also Figures S4–S6 and Table S1.



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leading to reduced levels of lipoylated proteins² and potentially limiting the amplification of the cuproptosis process. Our findings reveal that blocking PD-L1 to increase IFN- γ * T cell frequencies and IFN- γ -associated signaling enhances *FDX1* transcription, counteracting its depletion and amplifying tumor cuproptosis. This mechanism highlights the interplay between *FDX1* regulation, protein lipoylation, and immune-mediated augmentation of cuproptosis in tumors.

These two facets of *FDX1*—where *FDX1*-mediated protein lipoylation is both essential for cuproptosis and depleted during the process²—also serve different biomarker purposes: (1) predictive biomarker (baseline): tumors with high baseline *FDX1* expression or immunotherapy/IFN- γ -induced upregulation of *FDX1* may be intrinsically more susceptible to cuproptosis inducers such as elesclomol-copper; (2) pharmacodynamic marker (on-treatment): a reduction in *FDX1* protein following treatment with elesclomol-copper reflects active cuproptosis induction and therefore confirms the effectiveness of cuproptosis-inducing therapy. Further investigations are needed to translate these potential biomarkers into clinical applications.

Limitations of the study

Although our study emphasizes the IFN- γ -STAT1-IRF1-*FDX1* axis in regulating cuproptosis in cancer cells, we acknowledge that IFN- γ exerts pleiotropic effects on cellular metabolism (for example, the RNA-seq analysis also showed that IFN- γ increased expression of multiple genes encoding ETC subunits; Figure S6F). Therefore, additional IRF1-*FDX1*-independent mechanisms may also contribute to cuproptosis sensitization. However, our data demonstrate that the IFN- γ -STAT1-IRF1-*FDX1* pathway is at least one critical mechanism. Moreover, although our conclusions are supported by extensive functional, genetic, metabolic, and *in vivo* evidence, we acknowledge that future electron microscopy analysis would provide valuable additional ultrastructural insight into cuproptosis within our specific experimental context. In addition, the proposed combination of a cuproptosis inducer with immunotherapy for overcoming ICB resistance will require further validation in future clinical trials. We hope that our current study will serve as a catalyst for additional clinical investigations in this area.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for materials should be directed to the lead contact, Boyi Gan (bgan@mdanderson.org).

Materials availability

All materials used in this study are available from the lead contact with a completed materials transfer agreement.

Data and code availability

Polar metabolomics data generated in this study have been deposited at the NIH Common Fund's National Metabolomics Data Repository (NMDR) and the Metabolomics Workbench (Metabolomics Workbench: ST004558; data-track_id: 7011). RNA-seq data are provided in Table S1. Additional data are available from the lead contact upon reasonable request. This paper does not report original code.

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AUTHOR CONTRIBUTIONS

G.L. and B.G. conceived and designed the study and wrote the manuscript; G.L., Z. Lu, Z.X., and B.G. designed the experiments; Z. Lu and Z.X. performed most of the experiments with assistance from G.L., D.H., T.H., S.W., M.S., X. Zhao, Q.L., X.C., Y.Y., H.L., C.M., and L.Z.; C.B. and S.H.L. performed scRNA-seq; L.-T.K. and Z. Li analyzed scRNA-seq data; J.Y. analyzed ChIP-seq data from the ENCODE database; L.T. performed the metabolomic analysis under the guidance of E.G.; J.G. and C.L. performed RNAscope analyses; N.P., H.B., and J.W.W. generated and provided the anti-PD-1/PD-L1-resistant 344SQ_R cells; L.H., J.Z., H.T., J.-K.J.L., D.G., A.V., and J.H. generated and provided

Figure 6. Anti-PD-L1 therapy increases tumor sensitivity to cuproptosis induction

(A) PD-L1 staining scores in MC38 tumors from immunocompetent mice treated with elesclomol-CuCl₂.
(B) MC38 tumor volumes and weights in immunocompetent mice treated with elesclomol-CuCl₂ and/or anti-PD-L1.
(C and D) Tumor volumes and weights of MC38 (C) or orthotopic EMT6 (D) tumors in immunocompetent mice treated with elesclomol + copper(II) gluconate (CuGlu₂) and/or anti-PD-L1.
(E) Bliss independence synergy scores for anti-PD-L1 plus elesclomol + CuGlu₂ in the models shown in (C) and (D).
(F) PD-L1 staining scores in MC38 or orthotopic EMT6 tumors from immunocompetent mice treated with elesclomol + CuGlu₂.
(G and H) Proportion of IFN- γ * cells among CD8* (G) or CD4* (H) T cells in MC38 or orthotopic EMT6 tumors under indicated treatments.
(I–L) Representative images or scores of immunohistochemical staining for IRF1, *FDX1*, and lipoylated proteins (I), relative *FDX1* mRNA levels by advanced cell diagnostics (ACD) score (J), representative images or relative fluorescence intensity of Cu* staining (K), or representative images or relative DLAT foci counts (L) in MC38 tumors under indicated treatments. Scale bars, 20 μ m (I and K) and 40 μ m (L).
(M and N) Schematics of reciprocal crosstalk between cuproptotic tumor cells and immune cells (M) and of the synergistic antitumor mechanism of cuproptosis induction plus anti-PD-L1 therapy (N).
Data are mean $\pm$ SD, n = 6–8 tumors (B–D) or 5 tumor samples (G and H). (A, E, F, I, and J–L) Violin plots with median and IQR; synergy scores were calculated at 5 time points per model for (E); n = 5–8 random fields for (A), (F), (I), and (J)–(L). Unpaired two-tailed t test or two-way ANOVA; * p < 0.05, ** p < 0.01, *** p < 0.001, and **** p < 0.0001; ns, not significant.

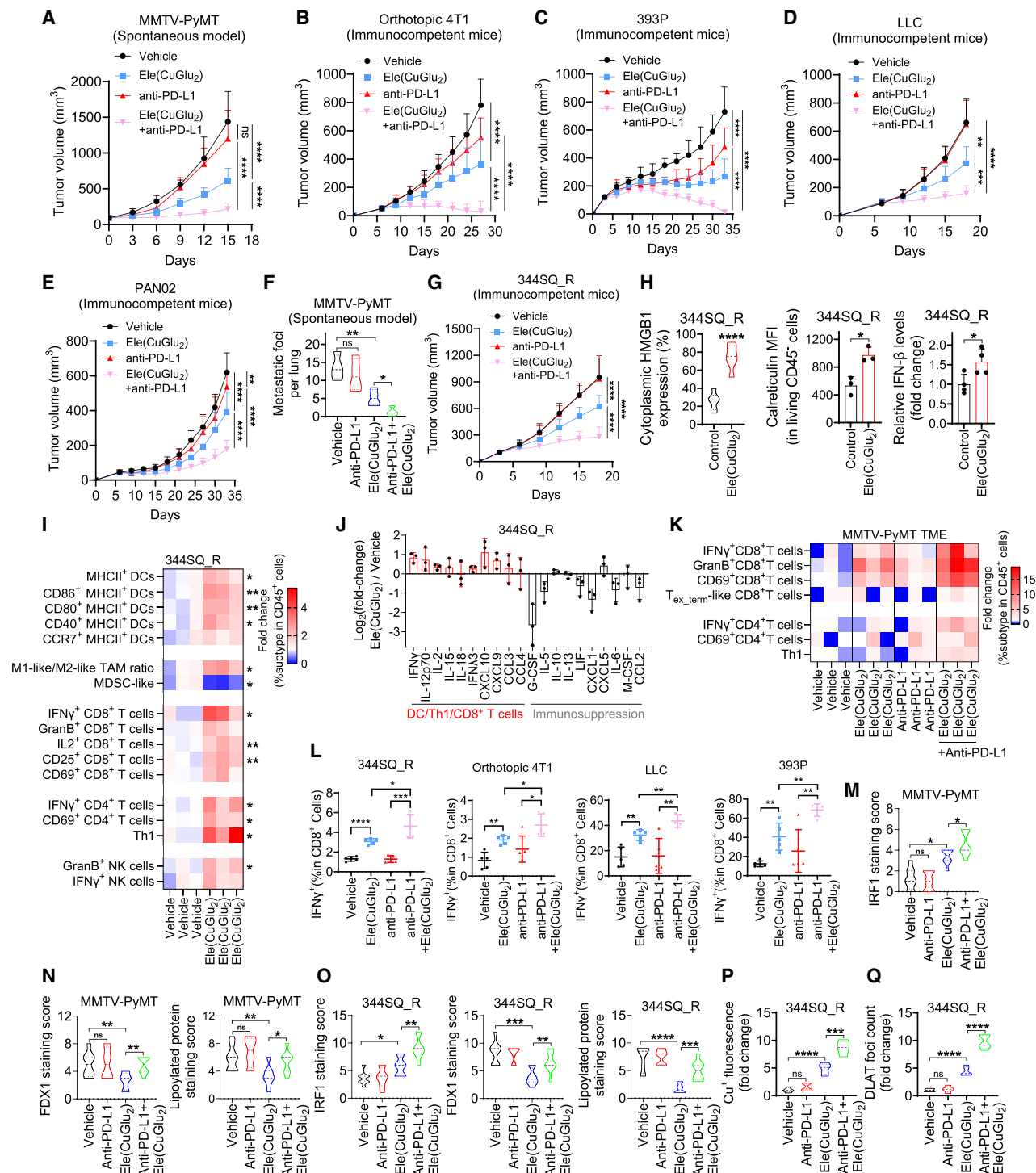


Figure 7. The cuproptosis inducer synergizes with immunotherapy to mitigate therapeutic resistance

(A–E) Tumor volumes over time in immunocompetent mice bearing MMTV-PyMT (A), orthotopic 4T1 (B), 393P (C), LLC (D), or PAN02 (E) tumors treated with elesclomol + CuGlu₂ and/or anti-PD-L1. For MMTV-PyMT, tumor volume denotes total tumor burden per mouse. (F) Metastatic foci in H&E-stained lung sections from MMTV-PyMT mice under indicated treatments. (G) Tumor volumes over time in immunocompetent mice bearing 344SQ-resistant (344SQ_R) tumors treated with elesclomol + CuGlu₂ and/or anti-PD-L1. (H) Percentage of cytoplasmic HMGB1 expression, calreticulin MFI on live CD45⁺ cells, or relative IFN-β levels in 344SQ_R tumors treated with elesclomol + CuGlu₂.

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baseline NSCLC patient samples; X. Zang provided immunological guidance; M.J.Y. performed histopathological analyses; B.G. supervised the project, established the collaboration, and provided funding support for the project; all authors commented on the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing financial interests.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

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 - Histology, immunohistochemistry, and RNAscope
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- **QUANTIFICATION AND STATISTICAL ANALYSIS**

SUPPLEMENTAL INFORMATION

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REFERENCES

1. Tang, D., Kroemer, G., and Kang, R. (2024). Targeting cuproptosis and cuproptosis in cancer. *Nat. Rev. Clin. Oncol.* 21, 370–388. <https://doi.org/10.1038/s41571-024-00876-0>.
2. Tsvetkov, P., Coy, S., Petrova, B., Dreishpoon, M., Verma, A., Abdusamad, M., Rossen, J., Joesch-Cohen, L., Humeidi, R., Spangler, R.D., et al. (2022). Copper induces cell death by targeting lipoylated TCA cycle proteins. *Science* 375, 1254–1261. <https://doi.org/10.1126/science.abf0529>.
3. Mao, C., Wang, M., Zhuang, L., and Gan, B. (2024). Metabolic cell death in cancer: ferroptosis, cuproptosis, disulfidptosis, and beyond. *Protein Cell* 15, 642–660. <https://doi.org/10.1093/procel/pwae003>.
4. Jiang, D., Zhuang, L., Koong, A.C., and Gan, B. (2025). Cuproptosis in cancer: from molecular mechanisms to therapeutic intervention. *Trends Cancer* 12, 275–286. <https://doi.org/10.1016/j.trecan.2025.12.002>.
5. Guo, B., Yang, F., Zhang, L., Zhao, Q., Wang, W., Yin, L., Chen, D., Wang, M., Han, S., Xiao, H., and Xing, N. (2023). Cuproptosis Induced by ROS Responsive Nanoparticles with Elesclomol and Copper Combined with αPD-L1 for Enhanced Cancer Immunotherapy. *Adv. Mater.* 35, e2212267. <https://doi.org/10.1002/adma.202212267>.
6. Du, Y., Zhao, X., He, F., Gong, H., Yang, J., Wu, L., Cui, X., Gai, S., Yang, P., and Lin, J. (2024). A Vacancy-Engineering Ferroelectric Nanomedicine for Cuproptosis/Apoptosis Co-Activated Immunotherapy. *Adv. Mater.* 36, e2403253. <https://doi.org/10.1002/adma.202403253>.
7. Guan, M., Cheng, K., Xie, X.T., Li, Y., Ma, M.W., Zhang, B., Chen, S., Chen, W., Liu, B., Fan, J.X., and Zhao, Y.D. (2024). Regulating copper homeostasis of tumor cells to promote cuproptosis for enhancing breast cancer immunotherapy. *Nat. Commun.* 15, 10060. <https://doi.org/10.1038/s41467-024-54469-7>.
8. Platanias, L.C. (2005). Mechanisms of type-I- and type-II-interferon-mediated signalling. *Nat. Rev. Immunol.* 5, 375–386. <https://doi.org/10.1038/nri1604>.
9. Wang, W., Green, M., Choi, J.E., Gijón, M., Kennedy, P.D., Johnson, J.K., Liao, P., Lang, X., Kryczek, I., Sell, A., et al. (2019). CD8⁺ T cells regulate tumour ferroptosis during cancer immunotherapy. *Nature* 569, 270–274. <https://doi.org/10.1038/s41586-019-1170-y>.
10. Liao, P., Wang, W., Wang, W., Kryczek, I., Li, X., Bian, Y., Sell, A., Wei, S., Grove, S., Johnson, J.K., et al. (2022). CD8⁺ T cells and fatty acids orchestrate tumor ferroptosis and immunity via ACSL4. *Cancer Cell* 40, 365–378. <https://doi.org/10.1016/j.ccell.2022.02.003>.
11. Galluzzi, L., Buqué, A., Kepp, O., Zitvogel, L., and Kroemer, G. (2017). Immunogenic cell death in cancer and infectious disease. *Nat. Rev. Immunol.* 17, 97–111. <https://doi.org/10.1038/nri.2016.107>.
12. Wculek, S.K., Cueto, F.J., Mujal, A.M., Melero, I., Krummel, M.F., and Sancho, D. (2020). Dendritic cells in cancer immunology and immunotherapy. *Nat. Rev. Immunol.* 20, 7–24. <https://doi.org/10.1038/s41577-019-0210-z>.
13. Janssens, S., Rennen, S., and Agostinis, P. (2024). Decoding immunogenic cell death from a dendritic cell perspective. *Immunol. Rev.* 321, 350–370. <https://doi.org/10.1111/imr.13301>.

(I and J) Fold changes in the frequencies of indicated immune cell subsets (I) or the levels of indicated cytokines (J) in the 344SQ_R tumors under indicated treatments.

(K) Fold changes in frequencies of indicated T cell subsets in the MMTV-PyMT TME under indicated treatments.

(L) Proportion of IFN-γ⁺ cells among CD8⁺ T cells in 344SQ_R, 4T1, LLC, or 393P tumors under indicated treatments.

(M–Q) Immunohistochemical staining scores for IRF1, FDX1, or lipoylated proteins in MMTV-PyMT (M and N) or 344SQ_R (O) tumors, relative Cu⁺ fluorescence intensity (P), or DLAT foci levels (Q) in 344SQ_R tumors under the indicated treatments.

Data are mean ± SD, n = 6 mice for (A) (total tumor burden per mouse), 6–8 tumors for (B)–(E) and (G), 3–4 tumor samples for (H) (middle/right) and (J), and 5 tumor samples for (L). (F, H [left], and M–Q) Violin plots with median and IQR; n = 4 mice for (F) and 5–8 random fields for (H) (left) and (M)–(Q). (I and K) Heatmaps (n = 3 tumor samples). Unpaired two-tailed t test or two-way ANOVA; *p < 0.05, **p < 0.01, ***p < 0.001, and ****p < 0.0001; ns, not significant.

See also [Figures S7](#) and [S8](#).

14. Moon, C.Y., Belabed, M., Park, M.D., Mattiuz, R., Puleston, D., and Merad, M. (2025). Dendritic cell maturation in cancer. *Nat. Rev. Cancer* 25, 225–248. <https://doi.org/10.1038/s41568-024-00787-3>.
15. Galluzzi, L., Vitale, I., Warren, S., Adjemian, S., Agostinis, P., Martinez, A.B., Chan, T.A., Coukos, G., Demaria, S., Deutsch, E., et al. (2020). Consensus guidelines for the definition, detection and interpretation of immunogenic cell death. *J. Immunother. Cancer* 8, e000337. <https://doi.org/10.1136/jitc-2019-000337>.
16. Masopust, D., Awasthi, A., Bosselut, R., Brooks, D.G., Buggert, M., Chamoto, K., Cui, W., Dong, C., Farber, D.L., Gebhardt, T., et al. (2026). Guidelines for T cell nomenclature. *Nat. Rev. Immunol.* 26, 298–313. <https://doi.org/10.1038/s41577-025-01238-2>.
17. Ries, C.H., Cannarile, M.A., Hoves, S., Benz, J., Wartha, K., Runza, V., Rey-Giraud, F., Pradel, L.P., Feuerhake, F., Klamann, I., et al. (2014). Targeting tumor-associated macrophages with anti-CSF-1R antibody reveals a strategy for cancer therapy. *Cancer Cell* 25, 846–859. <https://doi.org/10.1016/j.ccr.2014.05.016>.
18. Xie, Q., Ding, J., and Chen, Y. (2021). Role of CD8+ T lymphocyte cells: Interplay with stromal cells in tumor microenvironment. *Acta Pharm. Sin. B* 11, 1365–1378. <https://doi.org/10.1016/j.apsb.2021.03.027>.
19. Der, S.D., Zhou, A., Williams, B.R., and Silverman, R.H. (1998). Identification of genes differentially regulated by interferon alpha, beta, or gamma using oligonucleotide arrays. *Proc. Natl. Acad. Sci. USA* 95, 15623–15628. <https://doi.org/10.1073/pnas.95.26.15623>.
20. Rosain, J., Neehus, A.L., Manry, J., Yang, R., Le Pen, J., Daher, W., Liu, Z., Chan, Y.H., Tahuil, N., Türel, Ö., et al. (2023). Human IRF1 governs macrophagic IFN-gamma immunity to mycobacteria. *Cell* 186, 621–645. <https://doi.org/10.1016/j.cell.2022.12.038>.
21. Taniguchi, T., Ogasawara, K., Takaoka, A., and Tanaka, N. (2001). IRF family of transcription factors as regulators of host defense. *Annu. Rev. Immunol.* 19, 623–655. <https://doi.org/10.1146/annurev.immunol.19.1.623>.
22. Tamura, T., Yanai, H., Savitsky, D., and Taniguchi, T. (2008). The IRF family transcription factors in immunity and oncogenesis. *Annu. Rev. Immunol.* 26, 535–584. <https://doi.org/10.1146/annurev.immunol.26.021607.090400>.
23. Tsvetkov, P., Detappe, A., Cai, K., Keys, H.R., Brune, Z., Ying, W., Thiru, P., Reidy, M., Kugener, G., Rossen, J., et al. (2019). Mitochondrial metabolism promotes adaptation to proteotoxic stress. *Nat. Chem. Biol.* 15, 681–689. <https://doi.org/10.1038/s41589-019-0291-9>.
24. Liddle, F.J., Alvarez, J.V., Poli, V., and Frank, D.A. (2006). Tyrosine phosphorylation is required for functional activation of disulfide-containing constitutively active STAT mutants. *Biochemistry* 45, 5599–5605. <https://doi.org/10.1021/bi0525674>.
25. Park, J., Kim, K., Lee, E.J., Seo, Y.J., Lim, S.N., Park, K., Rho, S.B., Lee, S.H., and Lee, J.H. (2007). Elevated level of SUMOylated IRF-1 in tumor cells interferes with IRF-1-mediated apoptosis. *Proc. Natl. Acad. Sci. USA* 104, 17028–17033. <https://doi.org/10.1073/pnas.0609852104>.
26. Imamichi, Y., Mizutani, T., Ju, Y., Matsumura, T., Kawabe, S., Kanno, M., Yazawa, T., and Miyamoto, K. (2013). Transcriptional regulation of human ferredoxin 1 in ovarian granulosa cells. *Mol. Cell. Endocrinol.* 370, 1–10. <https://doi.org/10.1016/j.mce.2013.02.012>.
27. Sharma, P., Goswami, S., Raychaudhuri, D., Siddiqui, B.A., Singh, P., Nagarajan, A., Liu, J., Subudhi, S.K., Poon, C., Gant, K.L., et al. (2023). Immune checkpoint therapy-current perspectives and future directions. *Cell* 186, 1652–1669. <https://doi.org/10.1016/j.cell.2023.03.006>.
28. Voli, F., Valli, E., Lerra, L., Kimpton, K., Saletta, F., Giorgi, F.M., Mercatelli, D., Rouaen, J.R.C., Shen, S., Murray, J.E., et al. (2020). Intratumoral Copper Modulates PD-L1 Expression and Influences Tumor Immune Evasion. *Cancer Res.* 80, 4129–4144. <https://doi.org/10.1158/0008-5472.CAN-20-0471>.
29. Myint, Z.W., Oo, T.H., Thein, K.Z., Tun, A.M., and Saeed, H. (2018). Copper deficiency anemia: review article. *Ann. Hematol.* 97, 1527–1534. <https://doi.org/10.1007/s00277-018-3407-5>.
30. Sharma, P., Hu-Lieskovan, S., Wargo, J.A., and Ribas, A. (2017). Primary, Adaptive, and Acquired Resistance to Cancer Immunotherapy. *Cell* 168, 707–723. <https://doi.org/10.1016/j.cell.2017.01.017>.
31. Restifo, N.P., Smyth, M.J., and Snyder, A. (2016). Acquired resistance to immunotherapy and future challenges. *Nat. Rev. Cancer* 16, 121–126. <https://doi.org/10.1038/nrc.2016.2>.
32. Galassi, C., Chan, T.A., Vitale, I., and Galluzzi, L. (2024). The hallmarks of cancer immune evasion. *Cancer Cell* 42, 1825–1863. <https://doi.org/10.1016/j.ccell.2024.09.010>.
33. Wang, X., Schoenhals, J.E., Li, A., Valdecana, D.R., Ye, H., Zang, F., Tang, C., Tang, M., Liu, C.G., Liu, X., et al. (2017). Suppression of Type I IFN Signaling in Tumors Mediates Resistance to Anti-PD-1 Treatment That Can Be Overcome by Radiotherapy. *Cancer Res.* 77, 839–850. <https://doi.org/10.1158/0008-5472.CAN-15-3142>.
34. Zimmermannova, O., Ferreira, A.G., Ascic, E., Velasco Santiago, M., Kur-ochkin, I., Hansen, M., Met, Ö., Caiado, I., Shapiro, I.E., Michaux, J., et al. (2023). Restoring tumor immunogenicity with dendritic cell reprogramming. *Sci. Immunol.* 8, eadd4817. <https://doi.org/10.1126/sciimmunol.add4817>.
35. Baird, J.R., Friedman, D., Cottam, B., Dubensky, T.W., Jr., Kanne, D.B., Bambina, S., Bahjat, K., Crittenden, M.R., and Gough, M.J. (2016). Radiotherapy Combined with Novel STING-Targeting Oligonucleotides Results in Regression of Established Tumors. *Cancer Res.* 76, 50–61. <https://doi.org/10.1158/0008-5472.CAN-14-3619>.
36. Zhu, Y., Knolhoff, B.L., Meyer, M.A., Nywening, T.M., West, B.L., Luo, J., Wang-Gillam, A., Goedegebuure, S.P., Linehan, D.C., and DeNardo, D.G. (2014). CSF1/CSF1R blockade reprograms tumor-infiltrating macrophages and improves response to T-cell checkpoint immunotherapy in pancreatic cancer models. *Cancer Res.* 74, 5057–5069. <https://doi.org/10.1158/0008-5472.CAN-13-3723>.
37. Oba, T., Long, M.D., Keler, T., Marsh, H.C., Minderman, H., Abrams, S.I., Liu, S., and Ito, F. (2020). Overcoming primary and acquired resistance to anti-PD-L1 therapy by induction and activation of tumor-residing cDC1s. *Nat. Commun.* 11, 5415. <https://doi.org/10.1038/s41467-020-19192-z>.
38. Kim, K., Skora, A.D., Li, Z., Liu, Q., Tam, A.J., Blosser, R.L., Diaz, L.A., Jr., Papadopoulos, N., Kinzler, K.W., Vogelstein, B., and Zhou, S. (2014). Eradication of metastatic mouse cancers resistant to immune checkpoint blockade by suppression of myeloid-derived cells. *Proc. Natl. Acad. Sci. USA* 111, 11774–11779. <https://doi.org/10.1073/pnas.1410626111>.
39. Ouzounova, M., Lee, E., Piranlioglu, R., El Andaloussi, A., Kolhe, R., Demirci, M.F., Marasco, D., Asm, I., Chadli, A., Hassan, K.A., et al. (2017). Monocytic and granulocytic myeloid derived suppressor cells differentially regulate spatiotemporal tumour plasticity during metastatic cascade. *Nat. Commun.* 8, 14979. <https://doi.org/10.1038/ncomms14979>.
40. Kopecka, J., Salaroglio, I.C., Perez-Ruiz, E., Sarmiento-Ribeiro, A.B., Saponara, S., De Las Rivas, J., and Riganti, C. (2021). Hypoxia as a driver of resistance to immunotherapy. *Drug Resist. Updat.* 59, 100787. <https://doi.org/10.1016/j.drug.2021.100787>.
41. Yang, Z., Su, W., Wei, X., Pan, Y., Xing, M., Niu, L., Feng, B., Kong, W., Ren, X., Huang, F., et al. (2025). Hypoxia inducible factor-1alpha drives cancer resistance to cuproptosis. *Cancer Cell* 43, 937–954. <https://doi.org/10.1016/j.ccell.2025.02.015>.
42. Cappellesso, F., Orban, M.P., Shirgaonkar, N., Berardi, E., Serneels, J., Neveu, M.A., Di Molfetta, D., Piccapane, F., Caroppo, R., Debellis, L., et al. (2022). Targeting the bicarbonate transporter SLC4A4 overcomes immunosuppression and immunotherapy resistance in pancreatic cancer. *Nat. Cancer* 3, 1464–1483. <https://doi.org/10.1038/s43018-022-00470-2>.
43. Dixon, S.J., Lemberg, K.M., Lamprecht, M.R., Skouta, R., Zaitsev, E.M., Gleason, C.E., Patel, D.N., Bauer, A.J., Cantley, A.M., Yang, W.S., et al. (2012). Ferroptosis: an iron-dependent form of nonapoptotic cell death. *Cell* 149, 1060–1072. <https://doi.org/10.1016/j.cell.2012.03.042>.
44. Lei, G., Zhuang, L., and Gan, B. (2024). The roles of ferroptosis in cancer: Tumor suppression, tumor microenvironment, and therapeutic

- interventions. *Cancer Cell* 42, 513–534. <https://doi.org/10.1016/j.ccell.2024.03.011>.
45. Yan, C., Lv, H., Feng, Y., Li, Y., and Zhao, Z. (2024). Inhalable nanoparticles with enhanced cuproptosis and cGAS-STING activation for synergistic lung metastasis immunotherapy. *Acta Pharm. Sin. B* 14, 3697–3710. <https://doi.org/10.1016/j.apsb.2024.04.028>.
46. Zhu, C., Li, J., Sun, W., Li, D., Wang, Y., and Shen, X.C. (2024). Signaling Mechanism of Cuproptosis Activating cGAS-STING Immune Pathway. *JACS Au* 4, 3988–3999. <https://doi.org/10.1021/jacsau.4c00712>.
47. Wang, Y., Lu, J., Carisey, A.F., Chadchan, S.B., Lee, H.W., Malireddi, R.K.S., Sharma, B.R., Pandian, N., Tweedell, R.E., Palacios, G., et al. (2025). Innate immune and metabolic signals induce mitochondria-dependent membrane lysis via mitoyperiosis. *Cell* 188, 7155–7174. <https://doi.org/10.1016/j.cell.2025.11.002>.
48. Lei, G., Mao, C., Horbath, A.D., Yan, Y., Cai, S., Yao, J., Jiang, Y., Sun, M., Liu, X., Cheng, J., et al. (2024). BRCA1-Mediated Dual Regulation of Ferroptosis Exposes a Vulnerability to GPX4 and PARP Co-Inhibition in BRCA1-Deficient Cancers. *Cancer Discov.* 14, 1476–1495. <https://doi.org/10.1158/2159-8290.Cd-23-1220>.
49. Chauhan, A.S., Liu, X., Jing, J., Lee, H., Yadav, R.K., Liu, J., Zhou, Y., and Gan, B. (2019). STIM2 interacts with AMPK and regulates calcium-induced AMPK activation. *FASEB J.* 33, 2957–2970. <https://doi.org/10.1096/fj.201801225R>.
50. Lee, H., Zandkarimi, F., Zhang, Y., Meena, J.K., Kim, J., Zhuang, L., Tyagi, S., Ma, L., Westbrook, T.F., Steinberg, G.R., et al. (2020). Energy-stress-mediated AMPK activation inhibits ferroptosis. *Nat. Cell Biol.* 22, 225–234. <https://doi.org/10.1038/s41556-020-0461-8>.
51. Liu, X., and Gan, B. (2016). lncRNA NBR2 modulates cancer cell sensitivity to phenformin through GLUT1. *Cell Cycle (Georget. Tex.)* 15, 3471–3481. <https://doi.org/10.1080/15384101.2016.1249545>.
52. Koppula, P., Zhang, Y., Shi, J., Li, W., and Gan, B. (2017). The glutamate/cystine antiporter SLC7A11/xCT enhances cancer cell dependency on glucose by exporting glutamate. *J. Biol. Chem.* 292, 14240–14249. <https://doi.org/10.1074/jbc.M117.798405>.
53. Liu, X., Chen, Z., Yan, Y., Zandkarimi, F., Nie, L., Li, Q., Horbath, A., Olszewski, K., Kondiparthi, L., Mao, C., et al. (2024). Proteomic analysis of ferroptosis pathways reveals a role of CEPT1 in suppressing ferroptosis. *Protein Cell* 15, 686–703. <https://doi.org/10.1093/procel/pwae004>.
54. Koppula, P., Lei, G., Zhang, Y., Yan, Y., Mao, C., Kondiparthi, L., Shi, J., Liu, X., Horbath, A., Das, M., et al. (2022). A targetable CoQ-FSP1 axis drives ferroptosis- and radiation-resistance in KEAP1 inactive lung cancers. *Nat. Commun.* 13, 2206. <https://doi.org/10.1038/s41467-022-29905-1>.
55. Satija, R., Farrell, J.A., Gennert, D., Schier, A.F., and Regev, A. (2015). Spatial reconstruction of single-cell gene expression data. *Nat. Biotechnol.* 33, 495–502. <https://doi.org/10.1038/nbt.3192>.
56. Hou, W., and Ji, Z. (2024). Assessing GPT-4 for cell type annotation in single-cell RNA-seq analysis. *Cell. Methods* 21, 1462–1465. <https://doi.org/10.1038/s41592-024-02235-4>.
57. Purbey, P.K., Seo, J., Paul, M.K., Iwamoto, K.S., Daly, A.E., Feng, A.C., Champhekar, A.S., Langerman, J., Campbell, K.M., Schae, D., et al. (2024). Opposing tumor-cell-intrinsic and -extrinsic roles of the IRF1 transcription factor in antitumor immunity. *Cell Rep.* 43, 114289. <https://doi.org/10.1016/j.celrep.2024.114289>.
58. Korsunsky, I., Millard, N., Fan, J., Slowikowski, K., Zhang, F., Wei, K., Baglaenko, Y., Brenner, M., Loh, P.R., and Raychaudhuri, S. (2019). Fast, sensitive and accurate integration of single-cell data with Harmony. *Nat. Methods* 16, 1289–1296. <https://doi.org/10.1038/s41592-019-0619-0>.
59. Aran, D., Looney, A.P., Liu, L., Wu, E., Fong, V., Hsu, A., Chak, S., Naikawadi, R.P., Wolters, P.J., Abate, A.R., et al. (2019). Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage. *Nat. Immunol.* 20, 163–172. <https://doi.org/10.1038/s41590-018-0276-y>.
60. Shi, W., Wang, Y., Zhao, Y., Kim, J.J., Li, H., Meng, C., Chen, F., Zhang, J., Mak, D.H., Van, V., et al. (2023). Immune checkpoint B7-H3 is a therapeutic vulnerability in prostate cancer harboring PTEN and TP53 deficiencies. *Sci. Transl. Med.* 15, eadf6724. <https://doi.org/10.1126/scitranslmed.adf6724>.
61. Lee, H., Horbath, A., Kondiparthi, L., Meena, J.K., Lei, G., Dasgupta, S., Liu, X., Zhuang, L., Koppula, P., Li, M., et al. (2024). Cell cycle arrest induces lipid droplet formation and confers ferroptosis resistance. *Nat. Commun.* 15, 79. <https://doi.org/10.1038/s41467-023-44412-7>.
62. Lee, H., Dai, F., Zhuang, L., Xiao, Z.D., Kim, J., Zhang, Y., Ma, L., You, M.J., Wang, Z., and Gan, B. (2016). BAF180 regulates cellular senescence and hematopoietic stem cell homeostasis through p21. *Oncotarget* 7, 19134–19146. <https://doi.org/10.18632/oncotarget.8102>.
63. Koppula, P., Olszewski, K., Zhang, Y., Kondiparthi, L., Liu, X., Lei, G., Das, M., Fang, B., Poyurovsky, M.V., and Gan, B. (2021). KEAP1 deficiency drives glucose dependency and sensitizes lung cancer cells and tumors to GLUT inhibition. *iScience* 24, 102649. <https://doi.org/10.1016/j.isci.2021.102649>.
64. Wu, S., Mao, C., Kondiparthi, L., Poyurovsky, M.V., Olszewski, K., and Gan, B. (2022). A ferroptosis defense mechanism mediated by glycerol-3-phosphate dehydrogenase 2 in mitochondria. *Proc. Natl. Acad. Sci. USA* 119, e2121987119. <https://doi.org/10.1073/pnas.2121987119>.
65. Gan, B., Lim, C., Chu, G., Hua, S., Ding, Z., Collins, M., Hu, J., Jiang, S., Fletcher-Sanankone, E., Zhuang, L., et al. (2010). FoxOs enforce a progression checkpoint to constrain mTORC1-activated renal tumorigenesis. *Cancer Cell* 18, 472–484. <https://doi.org/10.1016/j.ccr.2010.10.019>.
66. Lin, A., Piao, H.L., Zhuang, L., Sarbassov dos, D., Ma, L., and Gan, B. (2014). FoxO transcription factors promote AKT Ser473 phosphorylation and renal tumor growth in response to pharmacologic inhibition of the PI3K-AKT pathway. *Cancer Res.* 74, 1682–1693. <https://doi.org/10.1158/0008-5472.CAN-13-1729>.
67. Lin, A., Yao, J., Zhuang, L., Wang, D., Han, J., Lam, E.W.-F., and TCGA Research Network, and Gan, B. (2014). The FoxO-BNIP3 axis exerts a unique regulation of mTORC1 and cell survival under energy stress. *Oncogene* 33, 3183–3194. <https://doi.org/10.1038/onc.2013.273>.
68. Zhang, Y., Shi, J., Liu, X., Feng, L., Gong, Z., Koppula, P., Sirohi, K., Li, X., Wei, Y., Lee, H., et al. (2018). BAP1 links metabolic regulation of ferroptosis to tumour suppression. *Nat. Cell Biol.* 20, 1181–1192. <https://doi.org/10.1038/s41556-018-0178-0>.
69. Tan, M., Cao, G., Wang, R., Cheng, L., Huang, W., Yin, Y., Ma, H., Ho, S.H., Wang, Z., Zhu, M., et al. (2024). Metal-ion-chelating phenylalanine nanostructures reverse immune dysfunction and sensitize breast tumour to immune checkpoint blockade. *Nat. Nanotechnol.* 19, 1903–1913. <https://doi.org/10.1038/s41565-024-01758-3>.
70. Li, J., Wu, C., Hu, H., Qin, G., Wu, X., Bai, F., Zhang, J., Cai, Y., Huang, Y., Wang, C., et al. (2023). Remodeling of the immune and stromal cell compartment by PD-1 blockade in mismatch repair-deficient colorectal cancer. *Cancer Cell* 41, 1152–1169. <https://doi.org/10.1016/j.ccell.2023.04.011>.
71. Necchi, A., Joseph, R.W., Loriot, Y., Hoffman-Censits, J., Perez-Gracia, J.L., Petrylak, D.P., Derleth, C.L., Tayama, D., Zhu, Q., Ding, B., et al. (2017). Atezolizumab in platinum-treated locally advanced or metastatic urothelial carcinoma: post-progression outcomes from the phase II IMvigor210 study. *Ann. Oncol.* 28, 3044–3050. <https://doi.org/10.1093/annonc/mdx518>.
72. Liu, D., Schilling, B., Liu, D., Sucker, A., Livingstone, E., Jerby-Amon, L., Zimmer, L., Gutzmer, R., Satzger, I., Loquai, C., et al. (2019). Integrative molecular and clinical modeling of clinical outcomes to PD1 blockade in patients with metastatic melanoma. *Nat. Med.* 25, 1916–1927. <https://doi.org/10.1038/s41591-019-0654-5>.
73. Braun, D.A., Hou, Y., Bakouny, Z., Ficial, M., Sant’Angelo, M., Forman, J., Ross-Macdonald, P., Berger, A.C., Jegede, O.A., Elagina, L., and Steinharter, J. (2020). Interplay of somatic alterations and immune infiltration modulates response to PD-1 blockade in advanced clear cell renal cell

- carcinoma. *Nat. Med.* 26, 909–918. <https://doi.org/10.1038/s41591-020-0839-y>.
74. Riaz, N., Havel, J.J., Makarov, V., Desrichard, A., Urba, W.J., Sims, J.S., Hodi, F.S., Martín-Algarra, S., Mandal, R., Sharfman, W.H., et al. (2017). Tumor and Microenvironment Evolution during Immunotherapy with Nivolumab. *Cell* 171, 934–949. <https://doi.org/10.1016/j.cell.2017.09.028>.
75. Kim, S.M., Park, Y.Y., Park, E.S., Cho, J.Y., Izzo, J.G., Zhang, D., Kim, S.B., Lee, J.H., Bhutani, M.S., Swisher, S.G., et al. (2010). Prognostic biomarkers for esophageal adenocarcinoma identified by analysis of tumor transcriptome. *PLoS One* 5, e15074. <https://doi.org/10.1371/journal.pone.0015074>.
76. Laibe, S., Lagarde, A., Ferrari, A., Monges, G., Birnbaum, D., and Olschwang, S.; COL2 Project (2012). A seven-gene signature aggregates a subgroup of stage II colon cancers with stage III. *Omics* 16, 560–565. <https://doi.org/10.1089/omi.2012.0039>.
77. Desmedt, C., Piette, F., Loi, S., Wang, Y., Lallemand, F., Haibe-Kains, B., Viale, G., Delorenzi, M., Zhang, Y., d'Assignies, M.S., et al. (2007). Strong time dependence of the 76-gene prognostic signature for node-negative breast cancer patients in the TRANSBIG multicenter independent validation series. *Clin. Cancer Res.* 13, 3207–3214. <https://doi.org/10.1158/1078-0432.CCR-06-2765>.
78. Mao, C., Lei, G., Horbath, A., Wang, M., Lu, Z., Yan, Y., Liu, X., Kondiparthi, L., Chen, X., Cheng, J., et al. (2024). Unraveling ETC complex I function in ferroptosis reveals a potential ferroptosis-inducing therapeutic strategy for LKB1-deficient cancers. *Mol. Cell* 84, 1964–1979. <https://doi.org/10.1016/j.molcel.2024.04.009>.
79. Yan, Y., Teng, H., Hang, Q., Kondiparthi, L., Lei, G., Horbath, A., Liu, X., Mao, C., Wu, S., Zhuang, L., et al. (2023). SLC7A11 expression level dictates differential responses to oxidative stress in cancer cells. *Nat. Commun.* 14, 3673. <https://doi.org/10.1038/s41467-023-39401-9>.
80. Fedchenko, N., and Reifennath, J. (2014). Different approaches for interpretation and reporting of immunohistochemistry analysis results in the bone tissue - a review. *Diagn. Pathol.* 9, 221. <https://doi.org/10.1186/s13000-014-0221-9>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-Lipoic Acid Rabbit pAb	Millipore	Cat#437695; RRID: AB_212120
DLAT (4A4-B6-C10) Mouse mAb	Cell Signaling Technology	Cat#12362; RRID: AB_2797893
FDX1 antibody	Abcam	Cat#ab108257; RRID: AB_10862209
LIAS antibody	Proteintech	Cat# 11577-1-AP; RRID: AB_2135972
MTF1 antibody	Proteintech	Cat#25383-1-AP; RRID: AB_2880053
IRF-1 (D5E4) XP Rabbit mAb	Cell Signaling Technology	Cat#8478; RRID: AB_10949108
STAT1 antibody	Cell Signaling Technology	Cat#14994; RRID: AB_2737027
Jak1 (6G4) Rabbit mAb	Cell Signaling Technology	Cat#3344; RRID: AB_2265054
Monoclonal Anti-Vinculin antibody produced in mouse	Sigma	Cat#V4505; RRID: AB_477617
β-Actin (8H10D10) Mouse mAb	Cell Signaling Technology	Cat#3700; RRID: AB_2242334
SDHB antibody	Proteintech	Cat#10620-1-AP; RRID: AB_2285522
PD-L1 antibody	Cell Signaling Technology	Cat#13684T; RRID: AB_2687655
HIF-1α antibody	Cell Signaling Technology	Cat#48085; RRID: AB_3073709
HMGB1 antibody	Abcam	Cat# ab18256; RRID: AB_444360
SF-1 antibody	Abclonal	Cat# A19277
Normal Rabbit IgG	Cell Signaling Technology	Cat#2729; RRID: AB_1031062
Calreticulin Antibody (681233) [PE]	Novus	Cat#FAB38981P; RRID: AB_3127218
InVivoMAb anti-mouse IFN-γ	BioXcell	Cat#BE0055;RRID: AB_1107694
InVivoMAb anti-human IFN-γ	BioXcell	Cat#BE0235;RRID: AB_2687717
InVivoMAb human IgG1 isotype control	BioXcell	Cat#BE0297; RRID: AB_2687817
InVivoMAb mouse IgG1 isotype control	BioXcell	Cat#BE0083; RRID: AB_1107784
InVivoMAb anti-mouse CD8α	BioXcell	Cat#BE0061;RRID: AB_1125541
InVivoMAb anti-mouse CD4	BioXcell	Cat#BE0003-1;RRID: AB_1107636
InVivoMAb anti-mouse NK1.1	BioXcell	Cat#BE0036;RRID: AB_1107737
InVivoPlus anti-mouse CSF1R (CD115)	BioXcell	Cat#BP0213;RRID: AB_2687699
InVivoSIM anti-human TNFα (Infliximab Biosimilar)	BioXcell	Cat#SIM0006;RRID: AB_2894727
InVivoMAb anti-mouse TNFα	BioXcell	Cat#BE0058;RRID: AB_1107764
InVivoMAb anti-mouse PD-L1	BioXcell	Cat#BE0101;RRID: AB_10949073
Mouse anti-human CD28	BD Biosciences	Cat#555726;RRID: AB_396069
Mouse anti-human CD3	BD Biosciences	Cat#550367;RRID: AB_393638
Rat anti-mouse CD16/CD32 (Fc block)	BD Biosciences	Cat#553142;RRID: AB_394657
PE anti-mouse CD45 Antibody	BioLegend	Cat#103106; RRID: AB_312971
APC/Cyanine7 anti-mouse CD45 Antibody	BioLegend	Cat#103116; RRID: AB_312981
PerCP/Cyanine5.5 anti-mouse CD3ε Antibody	BioLegend	Cat#100328;RRID: AB_893318
Alexa Fluor® 700 anti-mouse CD4 Antibody	BioLegend	Cat#100536;RRID: AB_493701
FITC anti-mouse CD8a Antibody	BioLegend	Cat#100706;RRID: AB_312745
PE anti-mouse IFN-γ Antibody	BioLegend	Cat#505808;RRID: AB_315402
PE anti-human/mouse Granzyme B Recombinant Antibody	BioLegend	Cat#372208;RRID: AB_2687032
APC anti-mouse Perforin Antibody	BioLegend	Cat#154404;RRID: AB_2721465
PerCP/Cyanine5.5 anti-mouse CD11c Antibody	BioLegend	Cat#117328;RRID: AB_2129641

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
FITC anti-mouse I-A/I-E (MHCII) Antibody	BioLegend	Cat#107606;RRID: AB_313321
PerCP/Cyanine5.5 anti-mouse/human CD11b Antibody	BioLegend	Cat#101228;RRID: AB_893232
FITC anti-mouse F4/80 Antibody	BioLegend	Cat#123108;RRID: AB_893502
Alexa Fluor® 700 anti-mouse CD86 Antibody	BioLegend	Cat#159214;RRID: AB_3106040
PE anti-mouse CD206 (MMR) Antibody	BioLegend	Cat#141706;RRID: AB_10895754
Anti-mouse H-2Db Adpgk Neopeptide Tetramer-ASMTNMELM-PE	MBL Life Science	Cat#TC-5113-1
OVA257-264 (SIINFEKL) peptide bound to H-2Kb mAb (eBio25-D1.16)-APC	ThermoFisher	Cat# 17-5743-80;RRID: AB_1311288
PE anti-mouse CD40 Antibody	Biolegend	Cat#157506;RRID: AB_2860731
PE/Cyanine7 anti-mouse CD80 Antibody	Biolegend	Cat# 104734;RRID: AB_2563112
APC anti-mouse CD274 (B7-H1, PD-L1) Antibody	Biolegend	Cat#124312;RRID: AB_10612741
Goat anti-Mouse IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 488	ThermoFisher	Cat# A-11029; RRID: AB_2534088
Goat anti-Rabbit IgG (H+L) Highly Cross-Adsorbed Secondary Antibody, Alexa Fluor 594	ThermoFisher	Cat# A-11037; RRID: AB_2534095
CyTOF antibody panel	This paper	Table S4

Biological samples

Baseline NSCLC tumor specimens	The University of Texas MD Anderson Cancer Center (MDACC)	N/A
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Chemicals, peptides, and recombinant proteins

Elesclomol	Selleckchem	Cat# S1052;CAS: 488832-69-5
Elesclomol	MedChemExpress	Cat# HY-12040;CAS: 488832-69-5
PX-478	MedChemExpress	Cat# HY-10231;CAS: 685898-44-6
Rotenone	MedChemExpress	Cat# HY-B1756;CAS: 83-79-4
Copper(II) chloride	Sigma	Cat# 203149;CAS: 7447-39-4
Copper(II) D-gluconate	Sigma	Cat# 344419;CAS: 527-09-3
Doxorubicin Hydrochloride	Selleckchem	Cat# S1208;CAS: 25316-40-9
Diphtheria toxin	Sigma	Cat# D0564
Antimycin A	Cayman	Cat# 19433;CAS: 642-15-9
Bathocuproinedisulfonic acid (BCS) disodium salt	Sigma	Cat# B1125;CAS: 52698-84-7
Ferrostatin-1	Sigma	Cat# SML0583;CAS: 347174-05-4
Necrostatin 2 racemate (Nec-1s)	Selleckchem	Cat# S8641;CAS: 852391-15-2
Z-VAD-fmk	Selleckchem	Cat# S7023;CAS: 187389-52-2
Mitoxantrone	Selleckchem	Cat# S2485;CAS: 70476-82-3
Mitomycin C	Sigma	Cat# M0503;CAS: 50-07-7
Lipofectamine® 2000 Transfection Reagent	ThermoFisher	Cat# 11668030
Propidium iodide	ThermoFisher	Cat# P3566
Cell-ID Cisplatin	Fluidigm	Cat# 201195
Recombinant mouse IFN-γ	R&D Systems	Cat# 485-MI-100
Recombinant human IFN-γ	R&D Systems	Cat# 285-IF-100
Recombinant human IL-2	ThermoFisher	Cat# 200-02-10UG
CopperSensor-1	TRC	Cat# TRC-B874740
Mouse M-CSF recombinant protein	ThermoFisher	Cat# 315-02-50UG
Mouse GM-CSF recombinant protein	ThermoFisher	Cat# 315-03-20UG
Matrigel® Basement Membrane Matrix	Corning	Cat# 354234

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Critical commercial assays		
PIPseq T100 3' Single Cell Capture and Lysis Kit	Fluent BioSciences	Cat#FB0004734
PIPseq T100 3' Single Cell Sequencing Kit	Fluent BioSciences	Cat#FB0004621
Mouse HMGB-1 ELISA Kit	Invitrogen	Cat# EEL102
Mouse IFN- β ELISA Kit, High Sensitivity	PBLAssaySCI	Cat# 42410
Extracellular Oxygen Consumption Assay Kit	Cayman	Cat# 502880
VECTASTAIN Elite ABC HRP Kit (Peroxidase, Rabbit IgG)	Vector Laboratory	Cat# PK-6101
VECTASTAIN® Elite® ABC-HRP Kit, Peroxidase (Mouse IgG)	Vector Laboratory	Cat# PK-6102
DAB Peroxidase (HRP) Substrate Kit (with Nickel), 3,3'-diaminobenzidine	Vector Laboratory	Cat# SK-4100
Pierce™ BCA Protein Assay Kit	ThermoFisher	Cat# 23227
RealTime-Glo™ Extracellular ATP Assay	Promega	Cat# GA5010
Vybrant™ Phagocytosis Assay Kit	ThermoFisher	Cat# V6694
Zombie Violet™ Fixable Viability Kit	BioLegend	Cat# 423114
BD Cytofix/Cytoperm Fixation/Permeabilization Kit	BD Biosciences	Cat# 554714
Q5® Site-Directed Mutagenesis Kit	NEB	Cat# E0554S
iScript Reverse Transcription Supermix	Bio-Rad	Cat# 1708841
SYBR GreenER qPCR SuperMix Universal	Invitrogen	Cat# 11762-100
SimpleChIP Enzymatic Chromatin IP Kit	Cell Signaling Technology	Cat# 9003
ChIP-Grade Protein G Magnetic Beads	Cell Signaling Technology	Cat# 9006
Dual-Glo Luciferase Assay System	Promega	Cat# E2920
RNAscope® 2.5 HD Brown Detection Kit	Advanced Cell Diagnostics (ACD)	Cat# 322300
RNAscope probe: mFDX1	Advanced Cell Diagnostics (ACD)	Cat# 1835431-C1
Macrophage Detachment Solution DXF	Sigma	Cat# C-41330
Deposited data		
Polar metabolomics data	This paper	Metabolomics Workbench: ST004558
Human urothelial carcinoma atezolizumab-treated cohort	Necchi et al. ⁷¹	IMvigor210; PMID: 28950298
Human metastatic melanoma anti-PD-1 WES/RNA-seq dataset	Liu et al. ⁷²	dbGaP: phs000452.v3.p1
Human advanced clear cell renal cell carcinoma PD-1 blockade cohort	Braun et al. ⁷³	N/A
Human melanoma nivolumab-treated RNA-seq dataset	Riaz et al. ⁷⁴	GEO: GSE91061
Human esophageal adenocarcinoma expression profiling dataset	Kim et al. ⁷⁵	GEO: GSE13898
Human non-small cell lung cancer expression profiling dataset	Gene Expression Omnibus (GEO)	GEO: GSE77803
Human stage II/III colon carcinoma expression profiling dataset	Laibe et al. ⁷⁶	GEO: GSE37892
Human node-negative breast cancer expression profiling dataset	Desmedt et al. ⁷⁷	GEO: GSE7390
Human d-MMR/MSI-H colorectal cancer anti-PD-1 scRNA-seq dataset	Li et al. ⁷⁰	GEO: GSE205506
K562 H3K4me3 ChIP-seq signal track	ENCODE Project	ENCODE: ENCSR000AKU
K562 H3K4me1 ChIP-seq signal track	ENCODE Project	ENCODE: ENCSR000AKS
K562 H3K27ac ChIP-seq signal track	ENCODE Project	ENCODE: ENCSR000AKP
K562 IRF1 ChIP-seq signal track, IFN- γ 30 min	ENCODE Project	ENCODE: ENCSR000EGK

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
K562 IRF1 ChIP-seq signal track, IFN- γ 6 h	ENCODE Project	ENCODE: ENCSR000EGT
K562 STAT1 ChIP-seq signal track, IFN- γ 30 min	ENCODE Project	ENCODE: ENCSR000EHK
K562 STAT1 ChIP-seq signal track, IFN- γ 6 h	ENCODE Project	ENCODE: ENCSR000EHJ
scRNA-seq in MC38 tumors	Purbey et al. ⁵⁷	GEO: GSE246923

Experimental models: Cell lines

393P	Steven H. Lin (MDACC)	N/A
MC38	Steven H. Lin (MDACC)	N/A
MC38-OVA	Steven H. Lin (MDACC)	N/A
Purified human CD8 ⁺ T cells	IQ Biosciences	Cat# IQB-Hu1-CD8T10
LKR13	Steven H. Lin (MDACC)	N/A
OE33	Steven H. Lin (MDACC)	N/A
HAP1	Junjie Chen (MDACC)	N/A
EMT6	Junjie Chen (MDACC)	N/A
4T1	Junjie Chen (MDACC)	N/A
LLC	Junjie Chen (MDACC)	N/A
PAN02	Haoqiang Ying (MDACC)	N/A
344SQ_R	James W. Welsh (MDACC)	N/A
MCF10A	ATCC	Cat# CRL-10317
HEK293T	ATCC	Cat# CRL-1573
Hs578T	ATCC	Cat# HTB-126

Experimental models: Organisms/strains

Mouse: athymic nude (Foxn1nu/Foxn1nu)	The Jackson Laboratory	RRID: IMSR_JAX:002019
Mouse: BALB/cJ	The Jackson Laboratory	RRID: IMSR_JAX:000651
Mouse: NOD scid gamma	The Jackson Laboratory	RRID: IMSR_JAX:005557
Mouse: C57BL/6J	The Jackson Laboratory	RRID: IMSR_JAX:000664
Mouse: 129S1/SvImJ	The Jackson Laboratory	RRID: IMSR_JAX:002448
Mouse: CD11c-DTR	The Jackson Laboratory	RRID: IMSR_JAX:004509
Mouse: MMTV-PyMT	The Jackson Laboratory	RRID: IMSR_JAX:002374

Oligonucleotides

sgRNA oligo sequences and primer sequences	This paper	Table S3
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Recombinant DNA

lentiCRISPR v2-puro	Addgene	Cat# 52961
lentiCRISPR v2-blast	Addgene	Cat# 83480
psPAX2	Addgene	Cat# 12260
pMD2.G	Addgene	Cat# 12259
pLX304	Addgene	Cat# 25890
pLX302	Addgene	Cat# 25896
pLV-EF1a-IRES-Puro	Addgene	Cat# 85132
pLV-EF1a-IRES-Blast	Addgene	Cat# 85133
pGL3-Basic	Addgene	Cat# 212936
pRL-TK	Promega	Cat# E2241
FDX1	DNASU	Clone ID: HsCD00514897
STAT1	Functional Genomics Core Facility (MDACC)	Cat#: OHS5893-202499266
IRF1	Functional Genomics Core Facility (MDACC)	Cat#: OHS5893-202494442
SF-1	Functional Genomics Core Facility (MDACC)	Cat#: OHS5894-202507236

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REAGENT or RESOURCE	SOURCE	IDENTIFIER
Software and algorithms		
GraphPad Prism 10	GraphPad	http://www.graphpad.com/
R v4.1.1	R Foundation for Statistical Computing	https://www.r-project.org/
Seurat v4.0.6	Satija et al. ⁵⁵	N/A
Harmony v0.1.0	Korsunsky et al. ⁵⁸	N/A
SingleR v1.8.0	Aran et al. ⁵⁹	N/A
GPTcelltype	Hou et al. ⁵⁶	N/A
Attune Flow Cytometer Software	ThermoFisher	https://www.thermofisher.com/us/en/home/life-science/cell-analysis/flow-cytometry/flow-cytometers/attune-nxt-flow-cytometer/software.html
Image Lab Software	Bio-Rad	https://www.bio-rad.com/en-us/product/image-lab-software?ID=KRE6P5E8Z
GSVA v1.46.0	Bioconductor	https://bioconductor.org/packages/GSVA/
inferCNV	Broad Institute	https://github.com/broadinstitute/inferCNV
ImageJ	ImageJ	https://imagej.net/ij/
Cytobank	Beckman Coulter	https://premium.cytobank.org
PIPseeker v3.3.0	Fluent BioSciences	https://www.fluentbio.com/products/pipseeker-software-for-data-analysis/
ZEN software	ZEISS	https://www.zeiss.com/microscopy/us/products/software/zeiss-zen.html
BioRender	BioRender	https://www.biorender.com/

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell culture studies

393P, MC38, MC38-OVA, LKR13, and OE33 cell lines were provided by Dr. Steven H. Lin at MD Anderson Cancer Center (MDACC). HAP1, EMT6, 4T1, and LLC cell lines were provided by Dr. Junjie Chen at MDACC. PAN02 cells were provided by Dr. Haoqiang Ying at MDACC. 344SQ_R cells were provided by Dr. James W. Welsh at MDACC. All other cancer cell lines were obtained from the American Type Culture Collection (ATCC). All cell lines were routinely tested for Mycoplasma contamination upon receipt. Cell line identities were authenticated by STR profiling where applicable or confirmed by the provider/vendor. Cell lines were banked in multiple aliquots upon receipt to minimize the risk of phenotypic drift, and all cell lines were cultured for less than one-month post-thaw. EMT6 cells were cultured in Waymouth's medium (Gibco) containing 15% (volume/volume; v/v) fetal bovine serum (Sigma) and 1% (v/v) penicillin/streptomycin (10,000 U/mL, Gibco). MCF10A cells were cultured in mammary epithelial cell growth medium BulletKit (MEGM; Lonza) with 100 ng/ml cholera toxin according to ATCC instructions. HAP1 cells were cultured in Iscove's modified Dulbecco's medium (IMDM, Gibco) with 10% (volume/volume; v/v) fetal bovine serum (Sigma) and 1% (v/v) penicillin/streptomycin (10,000 U/mL, Gibco). All other cell lines were cultured in RPMI-1640 (Gibco) or Dulbecco's modified Eagle medium (DMEM, Gibco) with 10% (volume/volume; v/v) fetal bovine serum (Sigma) and 1% (v/v) penicillin/streptomycin (10,000 U/mL, Gibco), at 37 °C with a humidified atmosphere of 20% O₂ and 5% CO₂, according to ATCC suggestions.

To prepare human CD8⁺ T-cell culture medium, 9 mL X-VIVO™ 15 serum-free hematopoietic cell medium (Lonza) was supplemented with 1 mL human serum (H3667, Sigma) (final 10% v/v), 50 U/mL recombinant human IL-2, 1 µg/mL mouse anti-human CD28 (BD), and 250 ng/mL mouse anti-human CD3 (BD). Purified human CD8⁺ T cells were purchased from IQ Biosciences (IQB-Hu1-CD8T10). Upon receipt, the cells were thawed, centrifuged at 300 × g for 5 min, resuspended in human CD8⁺ T-cell culture medium, and were cultured at 37°C with 5% CO₂. Collected supernatants were centrifuged at 10,000 × g for 10 min. The clarified supernatant was concentrated using 10-kDa MWCO centrifugal ultrafiltration units (3,000 × g for 15 min), filtered through a 0.22-µm membrane.

Animal models

Female 4- to 6-week-old athymic nude mice (*Foxn1*^{nu}/*Foxn1*^{nu}), BALB/cJ mice, NOD scid gamma mice, C57BL/6J mice, 129S1/SvImJ mice, CD11c-DTR mice, and FVB/N-Tg(MMTV-PyVT)634Mul/J mice (MMTV-PyMT) were obtained from the Jackson Laboratory. All mice were maintained under specific pathogen-free housing in the Animal Care Facility in the Department of Veterinary Medicine and Surgery (DVMS) at MD Anderson. Rodent housing conditions used in the current study were as follows: temperature set point 72 °F, high limit 74 °F, low limit 70 °F; humidity set point 45%, high limit 55%, low limit 40%; light cycle 12 hours light-dark. This study complied with all relevant ethical regulations regarding animal research in MD Anderson and was supervised by the DVMS throughout.

Orthotopic mammary fat pad implantation was conducted as previously described.⁴⁸ Briefly, 4×10^5 4T1 or 5×10^5 EMT6 cells were resuspended in 100 μ L of 50% volume Matrigel + 50% volume medium and implanted into the left fourth inguinal mammary fat pad of nude mice or BALB/cJ mice. For subcutaneous tumor implantation, tumor cells were inoculated in the left flank of each mouse after hair removal. In brief, 5×10^6 393P, 6×10^6 LKR13, or 5×10^5 344SQ_R cells were resuspended in 100 μ L of medium and injected into nude mice or 129S1/SvImJ mice subcutaneously; 1×10^6 MC38 (including wild-type, Cas9 control, *STAT1*-sgRNA-infected, *IRF1*-sgRNA-infected, and *FDX1*-sgRNA-infected cells), 1×10^6 LLC, or 1×10^6 PAN02 cells were resuspended in 100 μ L of medium and injected into nude mice or C57BL/6J mice subcutaneously. For human xenografts, 1×10^7 OE33 Cas9 control or *FDX1*-sgRNA-infected cells were resuspended in 100 μ L of 50% volume Matrigel + 50% volume medium and injected into NSG mice subcutaneously.

Patient samples

Baseline non-small cell lung cancer (NSCLC) tumor specimens were obtained from patients treated with immunotherapy at The University of Texas MD Anderson Cancer Center (Houston, TX, USA) through the GEMINI program. The procedures of human sample collection were approved by the Ethic Committee(s) of The University of Texas MD Anderson Cancer Center. Informed consent was obtained from all subjects. All available patient samples were evaluated by experienced pathologists for confirmation of histological type. Tumor response was assessed according to the Response Evaluation Criteria in Solid Tumors (RECIST). Patients with complete response (CR) or partial response (PR) were classified as responders, whereas patients with stable disease (SD) or progressive disease (PD) were classified as non-responders. Patient characteristics are provided in Table S2.

METHOD DETAILS

Constructs and CRISPR-Cas9-mediated gene knockout and overexpression cell line generation

Cell lines with knockout of target genes were generated as previously described.^{49,50} CRISPR-mediated KO plasmids containing gRNAs targeting *FDX1*, *IRF1*, *STAT1*, and *JAK1* were generated in lentiCRISPR v2 (Addgene) according to the standard protocol. The sequences of sgRNAs used in the current study are listed in Table S3. The resulting lentiCRISPR v2 sgRNA constructs were co-transfected into HEK293T cells with psPAX2 and pMD2.G packaging/envelope plasmids. Lentiviral supernatants were collected at 48–72 h after transfection, filtered, and used to transduce target cells in the presence of polybrene, followed by appropriate antibiotic selection. To delete the IRF1 binding motif on the human *FDX1* promoter, two flanking sgRNAs were cloned into lentiviral CRISPR vectors. Lentiviral particles were produced in HEK293T cells by co-transfecting the transfer plasmids with psPAX2 and pMD2.G, collected at 48–72 h, filtered, and used to transduce HAP1 cells in the presence of polybrene. Cells were selected with puromycin and blasticidin for 2–4 days to generate an antibiotic-resistant bulk population, followed by single-cell cloning by limiting dilution, expansion, and genotyping. Genomic DNA was PCR-amplified across the two cut sites to screen for the expected deletion band, and clones carrying the expected deletion were confirmed by sequencing across the junction.

FDX1, *STAT1*, and *IRF1* cDNAs were obtained from DNASU (*FDX1* clone ID: HsCD00514897) or the Functional Genomics Core Facility at MDACC and subcloned into lentiviral vectors. *FDX1* enzymatically inactive mutant, *STAT1*-C, and *IRF1*-K275,299R constructs were generated using the Q5® Site-Directed Mutagenesis Kit (NEB) according to the manufacturer's instructions. Stable cell lines overexpressing target gene constructs were generated as described previously.^{51,52} Briefly, HEK293T cells were co-transfected with empty vector or target gene constructs together with psPAX2 and pMD2.G. Viral supernatants were collected 48–72 h later, filtered, and used to transduce target cells in the presence of polybrene, followed by appropriate antibiotic selection.

Cell death and oxygen consumption assay

Cell death was measured by propidium iodide staining using a flow cytometer, as previously described.^{53,54} Briefly, cells were seeded onto 6- or 12-well plates and subsequently treated with CD8⁺ T cell-derived supernatants and/or indicated agents. Next, cells were collected (including floating dead cells) and stained with 5 μ g/mL propidium iodide. The percentage of the propidium iodide-positive dead cells was determined using the Attune NxT Flow Cytometer (ThermoFisher). A minimum of 5,000 single cells were analyzed per well. For oxygen consumption assay, cells were seeded in black clear-bottom 96-well plates and treated as indicated. Oxygen consumption rate was measured using the Extracellular Oxygen Consumption Assay Kit (Cayman Chemical, Item No. 502880) according to the manufacturer's instructions.

Cu⁺ probe fluorescence quantification

Intracellular and intratumoral Cu⁺ probe signals were assessed by CopperSensor-1 (CS1) fluorescence. Cells were fixed in 4% paraformaldehyde for 15 min, washed with PBS, incubated with CS1 (10 μ M) at 37 °C for 30 min in the dark, washed with PBS, counterstained with DAPI (1 μ g/mL, 5 min), washed with PBS, and mounted in glycerol-based mounting medium. For tissue CS1 staining, tumor sections were incubated with CS1 (10 μ M in PBS) at 37 °C for 30 min in the dark, washed with PBS, counterstained with DAPI (1 μ g/mL, 5 min, RT), rinsed with PBS, and mounted in glycerol-based mounting medium. Fluorescence was acquired on an LSM 880 microscope (Carl Zeiss, Oberkochen, Germany) using a 63 $\times$ oil-immersion objective and ZEN software, with identical acquisition settings across groups. Fluorescence intensity in multiple randomly selected fields was quantified using ImageJ.

Preparation of single-cell suspensions from tumors and lymph nodes

Fresh tumors, inguinal tumor-draining lymph nodes (TdLNs) and non-TdLNs were processed into single-cell suspensions. Tumors were trimmed to remove hair, skin, and necrotic regions, and tissues were mechanically dissociated using a gentleMACS™ Dissociator in DMEM-based dissociation buffer containing collagenase IV and DNase I (with BSA), following the manufacturer's recommendations. Cell suspensions were further treated with TrypLE™ Express (37°C, 5 min), filtered through 70- μ m strainers, and subjected to red blood cell lysis before counting. Cells were used immediately for downstream assays or cryopreserved in FBS containing 10% (v/v) DMSO.

Single-cell RNA sequencing (scRNA-seq) analysis

Cells were pelleted (500 $\times$ g, 5 min, 4°C), resuspended in staining buffer, and incubated with blocking buffer on ice (10 min). Each sample was then labeled with a unique hashing antibody (BioLegend) for 30 min at 4°C. After washing, cell pellets were pooled and processed using the PIPseq T100 3' Single Cell Capture and Lysis Kit and the PIPseq T100 3' Single Cell Sequencing Kit (Fluent BioSciences) according to the manufacturer's instructions. Briefly, 200,000 pooled cells were loaded into pre-templated instant partitions (PIPs) for single-cell mRNA capture, followed by cDNA synthesis/amplification and library construction. Libraries were quality-controlled by TapeStation (High Sensitivity D5000) and Qubit and/or NanoDrop, and sequenced by a service provider (Healgen Scientific, Houston, TX).

scRNA-seq data for 393P tumors were generated and pre-processed using PIPseeker (v3.3.0, Fluent BioScience), including demultiplexing, alignment, and gene count matrix generation. Quality control was performed for each sample using the Seurat package,⁵⁵ filtering cells based on the number of expressed genes, total UMI counts, and the proportion of mitochondrial gene-derived UMIs. No explicit batch correction was applied, as samples showed minimal batch effects upon inspection. Data normalization was applied using the "NormalizeData" function, followed by the identification of variable genes with "FindVariableFeatures." Data were scaled using the "ScaleData" function to center and standardize gene expression values across samples. Dimensionality reduction was carried out using principal component analysis (PCA) and uniform manifold approximation and projection (UMAP) with the "RunPCA" and "RunUMAP" functions, respectively. Cell clusters were then identified in the embedding space using "FindClusters". Cluster-specific marker genes were identified using the "FindAllMarkers" function in Seurat. These markers were used for cell type annotation. Preliminary cell-type annotation was assisted by GPTcelltype,⁵⁶ a Chat-GPT-powered cell type annotation tool, followed by manual curation based on canonical marker genes. Visualization included heatmaps and dot plots generated using Seurat functions ("DoHeatmap" and "DotPlot"), which display the average expression levels and the percentage of expressing cells for selected marker genes. Bar plots were used to summarize cell type proportions across different conditions.

scRNA-seq data for MC38 control and *IRF1*-deficient tumors from immunocompetent mice were downloaded from the GEO database (GEO: GSE246923)⁵⁷ and processed using Seurat (v4.0.6) in R (v4.1.1). Cells with fewer than 200 total UMI counts, fewer than 200 detected genes, or more than 20% mitochondrial transcripts were excluded. Data were normalized by scaling total UMIs per cell to 10,000 followed by log transformation, and the top 2,000 variable genes were identified using the variance-stabilizing transformation method. Batch effects were corrected using Harmony (v0.1.0)⁵⁸ with sample identity as the batch variable. Cells were visualized by tSNE using the top 50 principal components and clustered using FindClusters at a resolution of 0.4. Cluster marker genes were identified using FindAllMarkers, and clusters were annotated using the singleR (v1.8.0)⁵⁹ together with canonical marker genes. Malignant tumor cells were identified using inferCNV based on inferred copy-number variation profiles. Group-level average expression of selected cuproptosis-related genes was summarized using AverageExpression and displayed as fold changes in the heatmap. Differential-expression statistics were obtained using FindMarkers. For RNA-seq analysis in OE33 cells, gene-level FPKM values and read counts were generated for each sample. Heatmaps of selected genes were generated using FPKM-derived fold changes (or log2 fold changes as indicated) relative to the control group.

Analysis of polar metabolites by IC-HRMS

To determine the relative abundance of polar metabolites in cell samples, extracts were prepared and analyzed by ultra-high-resolution mass spectrometry (HRMS). Metabolites were extracted using ice-cold 50/30/20 (v/v) methanol/acetonitrile/water. Extracts were centrifuged at 17,000 g for 5 min at 4°C, and supernatants were transferred to clean tubes, followed by evaporation to dryness under nitrogen. Dried extracts were reconstituted in deionized water, and 10 μ L were injected for analysis by Ion chromatography (IC)-HRMS. IC mobile phase A (MPA; weak) was water, and mobile phase B (MPB; strong) was water containing 100 mM KOH. A Thermo Scientific Dionex ICS-6000+ system included a Thermo IonPac AS11 column (4 μ m particle size, 250 $\times$ 2 mm) with column

compartment kept at 35°C. The autosampler tray was chilled to 4°C. The mobile phase flow rate was 360 μ L/min, and the gradient elution program was: 0–2 min, 1% MPB; 2–25 min, 1–40% MPB; 25–39 min, 40–100% MPB; 39–50 min, 100% MPB; 50–50.5, 100–1% MPB. The total run time was 55 min. To enhance the desolvation for better sensitivity, methanol was delivered by an external pump and combined with the eluent via a low dead volume mixing tee. Data were acquired using a Thermo Orbitrap IQ-X Tribrid Mass Spectrometer under ESI negative ionization mode.

Flow cytometry and cytometry by time-of-flight (CyTOF) analysis of immune cells in tumors and lymph nodes

To perform FACS analysis, single cells from tumor samples were first incubated with anti-Mouse CD16/CD32 (BD Biosciences) for 5 min at 4 °C to block Fc receptors. Then cells were stained with CD45, CD3 ϵ , CD4 or CD8a antibodies (BioLegend) according to instructions. Zombie Violet™ Fixable Viability Kit (Biolegend) was also used to distinguish live cells from dead cells. Afterwards, cells were washed and treated with BD Cytfix/Cytoperm Fixation/Permeabilization Kit (BD Biosciences). The treated cells were then stained with IFN- γ , Granzyme B or Perforin antibodies (Biolegend). All samples were detected using Attune NxT Flow Cytometer (Invitrogen).

To perform FACS analysis, single cells from TdLNs were first incubated with anti-Mouse CD16/CD32 (BD Biosciences) for 5 min at 4 °C to block Fc receptors. To analyze antigen-specific priming, cells were stained with anti-mouse H-2Db Adpgk Neopeptide Tetramer-ASMTNMELM-PE (MBL life science) according to instructions. Afterwards, cells were stained with CD45, CD3 ϵ and CD8a (CD8⁺ T cell markers) antibodies (BioLegend) according to instructions. To analyze antigen cross presentation, cells were stained with OVA257-264 (SIINFEKL) peptide bound to H-2Kb Monoclonal Antibody (eBio25-D1.16 (25-D1.16))-APC (Invitrogen) according to instructions. Afterwards, cells were stained with CD45, CD11c and I-A/I-E (DC markers) antibodies (BioLegend) according to instructions. Zombie Violet™ Fixable Viability Kit (Biolegend) was also used to distinguish live cells from dead cells. All samples were detected using Attune NxT Flow Cytometer (Invitrogen).

CyTOF experiments were performed as described previously.⁶⁰ Single-cell suspensions from tumors or lymph nodes were prepared as described above. Cells were Fc-blocked (anti-CD16/32; BD Biosciences), labelled with Cell-ID Cisplatin (Fluidigm), and stained with metal-conjugated CyTOF surface antibody cocktails (30 min, 4°C). Cells were then fixed, permeabilized, and stained with intracellular antibody cocktails (1 h, RT). For DNA staining and singlet discrimination, cells were incubated in fixative containing Cell-ID Intercalator-Ir (Fluidigm) overnight at 4°C. Samples were acquired on a CyTOF® Helios™ mass cytometer. Antibodies are listed in Table S4.

Preparation of bone marrow-derived macrophages and dendritic cells

Bone marrow cells were isolated from femurs and tibias of healthy female C57BL/6J mice (5–6 weeks old) under aseptic conditions following CO₂ euthanasia. Cells were centrifuged at 400 $\times$ g for 5 min. Red blood cells were lysed by adding diluted Red Blood Cell Lysis Solution (10x) (Miltenyi Biotec Inc.). Cells were then washed once with complete DMEM and counted.

To generate unpolarized bone marrow-derived macrophages (BMDMs) (M0) and immature bone marrow-derived dendritic cells (iDCs), bone marrow cells were resuspended at 1 $\times$ 10⁶ cells/mL in complete DMEM supplemented with 20 ng/mL Mouse M-CSF Recombinant Protein (Fisher) (for M0) or 20 ng/mL Mouse GM-CSF Recombinant Protein (Fisher) (for iDCs), plated in low-attachment Petri dishes, and cultured at 37°C, 5% CO₂. On day 3, 50% of the supernatant was replaced with fresh cytokine-containing medium. For M0, cells were harvested using Macrophage Detachment Solution DXF (Sigma) and validated by flow cytometry using antibodies of macrophage surface markers (CD45, CD11b and F4/80, Biolegend) on day 6–7. For iDCs, semi-adherent cells were collected and assessed by flow cytometry using antibodies of DC markers (CD45, CD11c and I-A/I-E (MHC II), Biolegend).

Preparation of cuproptotic MC38 conditioned medium and assessment of macrophage polarization, DC maturation, and phagocytosis

MC38 cells (6 $\times$ 10⁵ per 10-cm dish) were treated with freshly prepared elesclomol–CuCl₂ in complete DMEM for a 2-h pulse, followed by replacement with fresh complete DMEM and incubation for an additional 24 h to allow cuproptosis to develop. Culture supernatants were collected and filtered through a 0.22- μ m membrane. At the time of conditioned-medium collection, viable cells were counted, and the control conditioned medium was diluted proportionally based on the relative number of viable cells.

To assess the effects of cuproptotic tumor cell conditioned medium on macrophage polarization and dendritic cell maturation, mouse M0 BMDMs or iDCs were seeded into 12-well plates at 5 $\times$ 10⁵ cells/well (1 mL/well). Then, 1 mL of cuproptotic tumor cell conditioned medium was added per well. Control wells received the same volume of conditioned medium from non-induced tumor cells. After 24 h, cells were harvested and stained for flow cytometry using antibodies of macrophage polarization markers, including CD45, CD11b, F4/80, CD86 and CD206 (MMR), or DC maturation/activation markers, including CD45, CD11c, I-A/I-E (MHC II), CD86, CD40, CD80, and PD-L1. All cells were stained according to the manufacturer's instructions. The stained cells were analyzed by flow cytometry.

To evaluate the effects of cuproptotic tumor cell conditioned medium on dendritic cell phagocytic capacity, mouse iDCs were seeded into black 96-well plates at 2 $\times$ 10⁴ cells/well (100 μ L per well). After overnight incubation, 100 μ L of cuproptotic tumor cell conditioned medium was added per well and incubated for 24 h. Equal volumes of fresh complete medium and conditioned medium from non-induced tumor cells were included as controls. Phagocytosis was assessed using the Vybrant™ Phagocytosis Assay Kit (Thermo) according to the manufacturer's instructions.

Assays for damage-associated molecular patterns (DAMPs) and cytokines

MC38, 393P, 4T1, or 344SQ_R cells were seeded into appropriate plates and cultured overnight. Cells were treated with elesclomol–CuCl₂ for a 2-h pulse, followed by replacement with fresh drug-free medium and incubation for the appropriate times to induce cuproptosis. Complete DMEM served as a negative control, and DMEM containing doxorubicin (DOX) served as a positive control. After the 2 h pulse, media were removed and replaced with fresh complete DMEM, and cells were cultured for an additional 24 h.

Extracellular ATP was quantified using RealTime-Glo™ Extracellular ATP Assay (Promega) following the manufacturer's protocol. For calreticulin exposure, cells were collected and stained with Zombie Violet™ Fixable Viability Kit (Biolegend) and Calreticulin Antibody (681233) [PE] (Novus) according to instructions, followed by flow cytometry analysis. For ELISA-based measurements, supernatants were collected and concentrated at 4°C. HMGB1 or IFN-β were quantified using Mouse HMGB-1 ELISA Kit (Invitrogen) or Mouse IFN-Beta ELISA Kit, High Sensitivity (PBLAssaySCI) according to the manufacturers' instructions.

For tumor calreticulin analysis, tumor single-cell suspensions were stained with CD45 and calreticulin antibodies, and CRT MFI was quantified in live CD45⁺ cells. For *in vivo* mouse cytokine measurements, tumors were homogenized on ice in cold DPBS, centrifuged at 10,000 × g for 10 min, and supernatants were stored at –80°C until analysis. IFN-β was quantified using Mouse IFN-Beta ELISA Kit, High Sensitivity (PBLAssaySCI), and other cytokines were measured using Luminex® 200™ platform.

Quantitative PCR with reverse transcription

Quantitative reverse transcription PCR was performed as previously described.^{61,62} Briefly, total RNA was extracted using TRIzol reagent (Invitrogen), and reverse transcription was performed using iScript Reverse Transcription Supermix (Biorad). SYBR GreenER qPCR SuperMix Universal (Invitrogen) was used for quantitative PCR in triplicate, with samples run on a Stratagene MX3000P qPCR system. The threshold cycle (C_t) values for each gene were normalized to those of β-actin, and the 2^{–ΔΔC_t} method was used for quantitative analysis. The primer sequences are listed in Table S3.

Western blot

Cell pellets were lysed using RIPA lysis buffer containing complete mini protease inhibitors (Roche).⁶³ Protein concentrations were detected by a Bicinchoninic Acid Protein Assay (Thermo Scientific) using a FLUOstar Omega microplate reader (BMG Labtech). Immunoprecipitation and Western blot analysis were conducted as previously described.^{54,64} The primary antibodies and concentrations used in the current study included the following: Lipic acid (1:1,000, 437695, Millipore), DLAT (1:1,000, 12362s, Cell Signaling), FDX1 (1:1,000, ab108257, Abcam), MTF1 (1:1,000, 25383-1-AP, Proteintech), LIAS (1:1,000, 11577-1-AP, Proteintech), SF-1 (1:500, A19277, Abclonal), IRF1 (1:1,000, 8478T, Cell Signaling), STAT1 (1:1,000, 14994, Cell Signaling), JAK1 (1:1,000, 3344T, Cell Signaling), Vinculin (1:10000; V4505, Sigma), and β-actin (1:5,000, 3700, Cell Signaling).

ChIP assays

ChIP assays were performed using the SimpleChIP Enzymatic Chromatin IP Kit (Cell Signaling, 9003) as previously described.^{65,66} Briefly, cells were fixed with formaldehyde (1% final volume concentration) for 10 minutes at room temperature. Fixation was stopped by adding glycine and incubating for 5 minutes at room temperature. Chromatin (10 μg) was incubated overnight with IRF1 and IgG control after digestion. Antibody-protein complex was captured with ChIP-Grade Protein G Magnetic Beads (Cell Signaling, 9006). ChIP DNA was analyzed by quantitative PCR with SYBR GreenER qPCR SuperMix Universal (Invitrogen) in a Stratagene MX3000P qPCR system using the primers. The primer sequences are listed in Table S3.

Luciferase reporter assay

Dual-luciferase assays were performed as described previously.^{67,68} The human *FDX1* promoter was cloned into pGL3-Basic. Cells—parental or IRF1-overexpressing as indicated—were co-transfected with the pGL3 constructs and pRL-TK (internal control) using Lipofectamine 2000 (ThermoFisher). After 48 h, cells were lysed, and firefly (F-luc) and Renilla (R-luc) activities were measured with the Dual-Glo Luciferase Assay System (Promega). Transcriptional activity was calculated as the F-luc/R-luc ratio.

Animal studies

In all animal studies, mice were randomized to different treatment groups when tumors reached 50 to 100 mm³. All consumables and reagents were sterile, and all preparation steps were performed in a sterile biosafety cabinet. To ensure accurate dosing, all formulations were freshly prepared immediately prior to use. For elesclomol–CuCl₂ treatment, elesclomol powder (100 mg) was fully dissolved in 1 mL DMSO. Separately, 9.5 mL PEG300 was mixed with 9.5 mL ddH₂O and then combined thoroughly with the elesclomol/DMSO solution to generate a clear, light-yellow stock at 5 mg/mL. Anhydrous CuCl₂ was dissolved in 1 mL ddH₂O to generate a 5 mg/mL stock solution. Immediately before dosing, elesclomol and CuCl₂ solutions were mixed at a ratio of 1 mL elesclomol solution plus 20 μL CuCl₂ solution to yield a clear deep-yellow solution, which was administered within 20 min. For mice with an average body weight of ~20 g, 0.1 mL was administered intraperitoneally (i.p.) per dose, corresponding to elesclomol 25 mg/kg and CuCl₂ 0.5 mg/kg.

For OE33 xenograft experiments, recombinant human IFN-γ was prepared in sterile PBS at 100 μg/mL and administered intraperitoneally at 0.5 mg/kg, corresponding to 10 μg in 100 μL for a 20-g mouse. IFN-γ was given three times per week, with each dose administered one day before elesclomol–CuCl₂ treatment.

For elesclomol-Copper(II) D-gluconate (CuGlu₂) treatment, CuGlu₂ (Sigma) was dissolved in ddH₂O. CuGlu₂ (12.5 mg/kg, oral administration), and Ele (25 mg/kg, i.p.) were administered to mice every 3 days until the endpoint as indicated in the corresponding figures. For treatment including antibodies, anti-mouse IFN- γ (250 μ g/mouse, i.p.), anti-mouse TNF α (200 μ g/mouse, i.p.), anti-mouse CD8 α (200 μ g/mouse, i.p.), anti-mouse CSF1R (CD115) (400 μ g/mouse, i.p.), anti-mouse CD4 (200 μ g/mouse, i.p.) or anti-mouse NK1.1 (300 μ g/mouse, i.p.) were given 1 day before treatment initiation then twice a week during treatment. Anti-mouse PD-L1 (100 μ g/mouse, i.p.) was given every 3 days during treatment. Equal amount of mouse IgG1 isotype control was added in the vehicle as control. Where indicated, bathocuproinedisulfonic acid (BCS) was co-administered (20 mg/kg, i.p.). The volume of tumors was measured with a caliper every 3 days until the endpoint and calculated according to the following equation: Volume = tumor length $\times$ tumor width² $\times$ 1/2 ($V = (L \times W^2)/2$).

For the spontaneous mammary tumor model, female MMTV-PyMT mice were monitored until total tumor burden reached $\sim$ 100 mm³, then treated with elesclomol (25 mg/kg, i.p.), anti-PD-L1 (100 μ g, i.p.), and CuGlu₂ (12.5 mg/kg, oral) every 3 days. Total tumor volume/weight was calculated as the sum of individual tumor volumes/weights per mouse. At endpoint, tumors and major organs were collected for histology; metastatic foci in lung sections were enumerated, and a subset of tumors was processed into single-cell suspensions for flow cytometry.

For PX-478 combination therapy, C57BL/6 mice bearing subcutaneous PAN02 tumors were treated starting on day 6 with elesclomol (25 mg/kg, i.p.), anti-PD-L1 (100 μ g, i.p.), CuGlu₂ (12.5 mg/kg, oral), and/or PX-478 (40 mg/kg, i.p.) every 3 days. For DC depletion, CD11c-DTR mice bearing subcutaneous MC38 tumors received diphtheria toxin (5 ng/g, i.p.) every 3 days throughout treatment to maintain depletion, as previously described,⁶⁹ and were treated with vehicle or elesclomol-CuCl₂ as used above. Non-depleted mice receiving vehicle or Ele(CuCl₂) were included as controls.

The combination effects were assessed with Bliss independence model and the score is defined as “observed combination effect” minus “expected additive effect”. The synergy indicates that the “observed combination effect” exceeds the “expected additive effect”.

Prophylactic vaccination assay

Female C57BL/6 mice (6–8 weeks old; 18–20 g; Jackson Laboratory) were randomized and maintained under SPF conditions. Mice were anesthetized and the dorsal area was shaved bilaterally. For vaccination, 1×10^6 MC38 cells undergoing (i) immunogenic apoptosis (mitoxantrone-induced), (ii) non-immunogenic apoptosis (mitomycin-C-induced), (iii) necrosis (three freeze-thaw cycles), or (iv) cuproptosis (Ele(CuCl₂)-induced) were injected subcutaneously into the left dorsal flank. One week later, mice were challenged by subcutaneous injection of 2×10^5 viable untreated MC38 cells into the contralateral (right) dorsal flank. Tumor growth at both inoculation sites was measured every 3 days.

Patient transcriptomic datasets analysis

To evaluate the association between FDX1 and immunotherapy outcomes, we analyzed a single-cell RNA-seq cohort (GEO: GSE205506),⁷⁰ retaining samples with >1,000 neoplastic cells and available immunotherapy annotations. We also analyzed four bulk transcriptomic immunotherapy cohorts with response and/or survival information: IMvigor210,⁷¹ dbGaP: phs000452.v3.p1,⁷² the David A. Braun cohort,⁷³ and GEO: GSE91061,⁷⁴ obtained from GEO, dbGaP, or the IMvigor210CoreBiologies R package (v1.0.0). Clinical response was categorized as responders (R; CR/PR) or non-responders (NR; SD/PD).

To assess associations between FDX1 expression and immune-related programs (STAT1/IRF1, ISGs, MHC-related molecules, and CD8⁺ T-cell activation signatures) across cancers, we used TCGA pan-cancer data together with additional GEO datasets representing major tumor types, including ESCA (GEO: GSE13898), lung (GEO: GSE77803), colon (GEO: GSE37892), and breast (GEO: GSE7390).^{75–77} Spearman’s correlation was used for association analyses, and CD8⁺ T-cell activation signatures were quantified by GSVA (GSVA package v1.46.0).

Histology, immunohistochemistry, and RNAscope

Histologic analysis and immunohistochemistry staining were performed as previously described.^{78,79} Briefly, tumor tissues and major organs were collected, immediately fixed in 10% neutral-buffered formalin (ThermoFisher) overnight, and stored in 70% ethanol at 4 °C. The tissues were dehydrated and embedded in paraffin by the Research Histology Core Laboratory at MDACC according to standard protocols. Samples were sectioned at a thickness of 5 μ m and subjected to hematoxylin and eosin or immunohistochemistry staining. Tissue microarray sections were obtained from TissueArray. The primary antibodies and concentrations used for immunohistochemistry included the following: Lipoic acid (1:600, 437695, Millipore), FDX1 (1:250, ab108257, Abcam), IRF1 (1:100, 8478T, Cell Signaling), STAT1 (1:1,600, 14994, Cell Signaling), PD-L1 (1:200, 13684T, Cell Signaling), HIF-1 α (1:200, 48085, Cell Signaling), and SDHB (1:200, 10620-1-AP, Proteintech). Staining was performed using the Vectastain elite ABC kit and DAB peroxidase substrate kit (Vector laboratories). Images were randomly acquired at 200 $\times$ or 400 $\times$ magnification using an Olympus BX43 microscope, and an immunoreactive score system was used to semi-quantitatively assess the level of staining.⁸⁰

RNAscope in situ hybridization was performed using the RNAscope® 2.5 HD Brown Detection Kit (Advanced Cell Diagnostics, ACD, Newark, CA) according to the manufacturer’s instructions. Briefly, formalin-fixed paraffin-embedded (FFPE) tissue sections were deparaffinized, treated with hydrogen peroxide for 10 min, subjected to target retrieval at 99 °C for 15 min, and incubated with Protease Plus at 40 °C for 30 min in a HybEZ™ oven. Then target probes specific to *mFDX1* (#1835431-C1, ACD) were

hybridized for 2 h at 40 °C, followed by a standard amplification step (AMP1–AMP6) and chromogenic development with DAB (3,3'-diaminobenzidine). Slides were counterstained with hematoxylin, dehydrated, and mounted with mounting medium (Cytoseal 60, Epredia). Images were captured using a bright-field microscope (Olympus BX43). RNA signals were quantified as puncta per cell using the ACD semi-quantitative scoring system (0–4 scores).

Immunofluorescence staining

For cellular DLAT staining, cells were fixed with 4% paraformaldehyde for 15 min, permeabilized with 0.5% Triton X-100 for 10 min, blocked, and incubated with anti-DLAT, followed by incubation with an Alexa Fluor 488–conjugated secondary antibody for 1 h at room temperature. Nuclei were counterstained with Hoechst. For tissue immunofluorescence staining, paraffin-embedded sections were deparaffinized, subjected to antigen retrieval, blocked with Background Sniper (Biocare Medical, BS966) for 20 min at room temperature, and incubated overnight at 4 °C with anti-DLAT or anti-HMGB1. After washing, sections were incubated with the appropriate Alexa Fluor 488– or 594–conjugated secondary antibody for 1 h, washed with PBS and PBST (0.1% Tween-20), counterstained with DAPI, and mounted. Fluorescence images were acquired on a Zeiss LSM 880 confocal microscope (63× oil) with identical settings across groups. DLAT foci were counted in multiple randomly selected fields. Cytoplasmic HMGB1 expression was quantified as the percentage of cells within tumor regions showing cytoplasmic HMGB1 signal among DAPI-positive cells.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses were performed with GraphPad Prism or R software using two-tailed unpaired Student's *t* test, two-way ANOVA, Wilcoxon rank-sum test, log-rank test, chi-squared test, Spearman's rank correlation, or Pearson's correlation, as indicated. Statistical analyses for scRNA-seq and bulk RNA-seq data were as described above. Data are presented as mean ± standard deviation (SD), violin plots with median and interquartile range (IQR), heatmaps, scatter plots with correlation analyses, or Kaplan–Meier survival curves, as indicated. Statistical significance levels (*p* values) are indicated as follows: **p* < 0.05; ***p* < 0.01; ****p* < 0.001; *****p* < 0.0001; ns, not significant. Sample sizes are indicated in the figure legends. No statistical methods were used to predetermine sample size.